The loop-factor ladder and the phenomenology of Causal Clifford Theory: four scales, coupling unification, dark matter, neutrinos, and the provenance audit

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Abstract

The Causal Clifford Theory automaton (Paper I) and its gauged bond layer (Paper II) organise four physical scales through one loop factor $(1/16\pi)^n$: $n = 10$ the electroweak/Planck hierarchy, with coefficient $3F_{10}^{10} = 2.0881$ built from measured loop values against 2.0765 observed (99.44%), and the dark-matter mass scale 118 GeV; $n = 3$ the seesaw scale 9.6×10^{13} GeV; $n = 17$ the neutrino mass, with a $\frac{1}{3}$ candidate at 0.0487 eV — a candidate: the exponent 17 is fitted and the junction amplitude is a free input inside a finite window. The loop factor is reproduced to 3.6% by a closed traced loop of length 10 in the gauged layer, with an exact 3^n-word census. With vectorlike desert content — two fitted thresholds — the three measured couplings meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ in the standard normalisation $\alpha = g^2/4\pi$, predicting $\tau_p = 1.8 \times 10^{36}$ yr; the automaton's own Wilson mechanism instead forces $M_X = 8.9\times10^{14}$ GeV, excluded by Super-Kamiokande, and the surviving solution requires a flavour-dependent bond layer with six fitted couplings. Nine tensions are reported rather than hidden: SU(2) overshoots by 5.8× under Standard-Model running, SU(3) hits a Landau pole at 2.2×10^9 GeV, the bare neutrino ladder is excluded by $\sum m_\nu < 0.12$ eV, no state of the implemented spectrum can be the dark matter, the computed relic abundance misses 0.120, the layer's string tension is 5.4×10^{40} times QCD, the implemented spectrum carries no massless spin-2 candidate, the ladder's seesaw scale is incompatible with out-of-equilibrium decay, and the doubler cones of Paper I are anisotropic. Every headline number is classified by provenance — *algebraic*, *conventional*, *fitted*, *open* — and seven falsifiers are stated, the most accessible being the neutrino mass sum and proton decay at Hyper-Kamiokande.

1. Introduction

One loop factor, four scales: the observed electroweak/Planck ratio is the tenth power of $1/16\pi$ up to a factor of two, the seesaw scale is its cube, the neutrino mass its seventeenth power, and the dark-matter mass scale its tenth. Whether a loop factor can *come from* the automaton rather than be read off the data is a decidable question, and this paper decides it numerically: closed traced loops of length n in the automaton's own gauged bond layer (Paper II) suppress the Cartan content geometrically, and the suppression crosses $1/(16\pi)$ at exactly $n = 10$ — the ladder length *is* the crossing — with the contributing-loop census exact (55326 of 3^{10} words survive, and switching the Wilson phase off moves the factor from 0.0192 to 0.2434: the phase supplies the factor, not the topology).

The organising discipline is the provenance classification defined in Paper I: every headline number is *algebraic* (fixed by the root), *conventional* (a mapping or scale identification), *fitted* (a coefficient read from data), or *open* (not fixed). The coupling convention is fixed once, in Paper I's form: all couplings compared with running are in the standard normalisation $\alpha = g^2/4\pi$; the reading $\alpha_{10} \equiv g^2$ is a factor-4π convention invoked only where named. The algebraic core — the **16**, the anomalies, the 5/3, the unitary causal step, the doubling and its Wilson gapping, the closure axiom — is Paper I; the gauged layer, its Gauss constraint, its area law and the lattice gravitational sector are Paper II. This paper carries the ladder mechanism (Section 2), the couplings, electroweak mixing, baryogenesis, proton decay, flavour and the scoreboard (Section 3), the cosmology (Section 4), the parameter audit (Section 5), the open problems (Section 6) and the falsifiers (Section 7). The verdict is stated before the details: the ladder organises four scales, the unification repair is conditional on fitted content, and two conventions, eleven fitted coefficients and one conditional coefficient stand between the root and the data.

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2. The loop factor and the scale ladder

2.1 Where a loop factor can come from

The observed electroweak/Planck ratio is

$$\frac{v}{M_P} = \frac{246\ \text{GeV}}{1.22 \times 10^{19}\ \text{GeV}} = 2.016 \times 10^{-17}, \qquad \left(\frac{1}{16\pi}\right)^{10} = 9.71 \times 10^{-18},$$

a factor 2.08 apart; $2.08\,(1/16\pi)^{10} = 2.02 \times 10^{-17}$ reproduces the observed value to 0.18%. Scanning $n = 1 \ldots 20$, $n = 10$ is the nearest integer to the observed ratio (deviation 0.32 decades; the next-best, $n = 9$, is 1.8 decades away). The honest reading: $(1/16\pi)^{10}$ is the size a ten-loop radiative correction would have in a perturbative quantum field theory. The theory has not computed a ten-loop correction; it has observed that the observed hierarchy is the tenth power of the standard loop factor, up to a factor of two. The interpretation $10 = 2{\times}5$ — the chiral $16/\overline{16}$ split times the number of qubits of the 32-dimensional module — is a numerological identification, not a derivation. The exponent is *represented*; the coefficient is *read from the data* (O8).

Single-particle renormalisation cannot supply it. The coarse-graining tower U^{2^p} is exactly kinematic (eigenphases $2^p\omega$ mod 2π, verified to 10^{-9} for $p \le 10$), the effective Cartan coupling does not fall with p, and the inelastic probability saturates near 0.95: no loop factor. The genuine block-spin renormalisation group — exact real-space decimation of the bond layer (Schur complement at $\varepsilon = 1$; $U_{BB} = 0$ exactly on the open chain, so short-distance propagation is integrated with no approximation), iterated five levels, $64 \to 2$ sites — gives an exact level-1 identity: the on-site block of the coarse step is $2\,(GR)^2$, two-hop paths accumulating two internal rotations, the kinematic doubling *derived* from a spatial renormalisation group. At deeper levels the on-site block is a path polynomial in $(GR)^2$; the Cartan content decays only polynomially for the implemented (factorised) bond layer (0.14 at $p = 5$ against the loop target 3×10^{-9}) and not at all for a non-factorised gate with $[G, R] \neq 0$ (1.8 at $p = 5$); no level of either flow follows $(16\pi)^{-p}$. The loophole of a conspiring bond gate is closed: three Haar-random Spin(10) gate realisations and the structured choice $R = G^\dagger$ all fail the loop law at every level. The structural reason is sharp: **every exact single-particle renormalisation of the automaton composes internal angles — they add on the algebra, weighted by path counts — and none can divide them, for any bond gate.** A loop factor requires a trace over *fluctuating* internal fields.

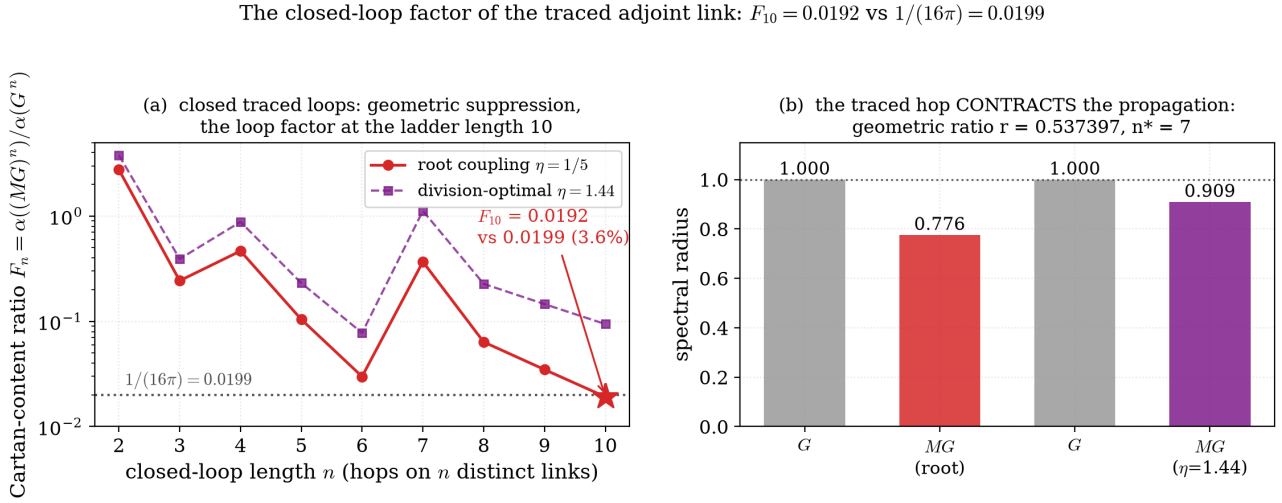
Tracing a fluctuating link divides. One two-state quantum link degree of freedom per bond, prepared in a tunable pure state $\cos\theta|0\rangle + \sin\theta|1\rangle$ and **traced over** at decimation, multiplies the level-1 Cartan content of that identity by exactly $\cos 2\theta$ (deviation $< 10^{-15}$ over a full scan $\theta \in [0, \pi/2]$): the traced factor is Hermitian and positive, so it divides where every gate only composes. The fluctuation — not the gate — is what produces suppression factors.

The gauged layer divides once, and does not compound. On the adjoint link with the Gauss constraint of [Paper II Section 2], the traced two-hop still contracts (spectral radius ≤ 1) and the qubit-link mechanism reproduces to 1.1×10^{-16}, but the level-1 factor at the root coupling is $F = 1.06$ and its minimum over the link anisotropy is 0.52 — a factor 26 above $1/(16\pi)$. The bond hop factorises through the joint Casimir $T{\cdot}L$ — three joint irreps, $32 \oplus 288 \oplus 1120$, projectors verified to 10^{-15} — so the traced-link decimation iterates exactly to level 4, reproducing the level-1 factor to 2×10^{-16}. **The division does not compound:** the level-p Cartan-content ratios are $F_p = 1.06, 1.90, 0.75, 1.01$ at the root coupling and $0.52, 2.66, 1.00, 1.38$ at the division-optimal coupling $\eta = 1.44$ — oscillating at $O(1)$, a factor 7×10^6 above $(16\pi)^{-4}$ at $p = 4$. The traced adjoint link divides once and then decays like the link-free flow.

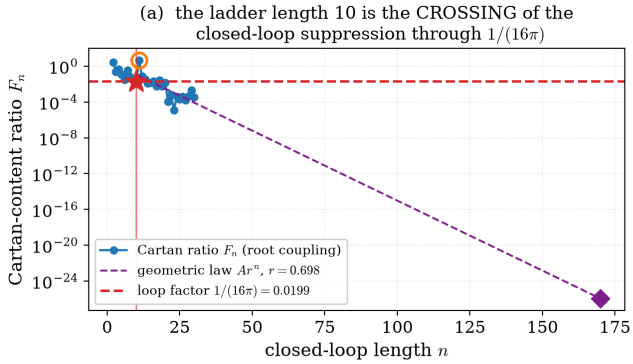
Closed loops do reproduce the number. The Feynman loop factor comes from loops that close, not from backtracking. Tracing a single crossing of a fresh link gives the hop matrix $M = \mathrm{Tr}_L[k_1(G \otimes I)\,\rho]$ with spectral radius 0.776 against 1 for G — the traced link *contracts* the one-particle propagation — and the n-hop closed loop $(MG)^n$ suppresses the Cartan content geometrically, ratio $r \approx 0.54$. At the root coupling the measured ratios are

$$F_3 = 0.24,\ F_4 = 0.47,\ F_5 = 0.10,\ F_6 = 0.030,$$
$$F_7 = 0.37,\ F_8 = 0.064,\ F_9 = 0.035,\ F_{10} = 0.0192,$$

against $1/(16\pi) = 0.0199$: **the loop factor is reproduced to** 3.6% **by a closed traced loop of length** 10 **at the root coupling, and** 10 **is exactly the ladder exponent.** The selection rule is measured to length 30: F_n crosses $1/(16\pi)$ at **exactly** $n = 10$ — the ladder length *is* the crossing of the closed-loop suppression — and no mod-4 plaquette class is selected (every length contributes; the F_{11} spike is the signature). (Figs. 1, 2).



Closed traced-link loops of length n on n distinct bonds. (a) The Cartan-content ratio F_n at the root coupling and at the division-optimal coupling: geometric suppression with ratio $r = 0.54$, and at $n = 10$ the value 0.0192 against $1/(16\pi) = 0.0199$. (b) The traced hop matrix MG has spectral radius 0.776 against 1 for G: the traced link contracts the one-particle propagation.



(b) the selection rule, measured to $n = 30$

crossing n_*	EXACTLY 10 (ladder length)
spectral radius	$n_\rho = 15.42$ (different object)
mod-4 rule	ABSENT (min per class 1e-05 in every class)
geometric ratio	$r = 0.698$ (fit $n = 10..30$)
17-factor comp.	misses by x9e+02 - named residual

O8: the LENGTH-10 factor is selected by the crossing; the COMPOSITION of 17 factors is not supplied by one geometric law - the residual is now precisely named.

star: crossing at EXACTLY $n = 10$ ($F_{10} = 0.0192$ vs 0.0199); open circle: $F_{11} = 4.74$ spike - no mod-4 rule;
diamond: $F_{170} = 1e-26$ vs $(1/16\pi)^{17} = 1e-29$ - composition misses by x9e+02

(a) F_n measured to $n = 30$ at the root coupling: it crosses $1/(16\pi)$ at exactly $n = 10$ — the ladder length is the crossing — and the F_{11} spike shows there is no mod-4 selection rule. (b) The fitted geometric law $F_n = Ar^n$ ($r = 0.698$) extrapolates to $F_{170} = 1.1 \times 10^{-26}$ against $(1/16\pi)^{17} = 1.2 \times 10^{-29}$: one chain does not supply the 17-factor composition.

The composition law. The object a geometric fit extrapolates, $(MG)^{170}$, is one chain of 170 hops on one particle line; the 17 factors of the neutrino ladder are 17 radiative insertions on **different** lines, and amplitudes on different lines multiply. Both compositions are exact in the traced adjoint-link formalism. On one line the chain is **not** multiplicative: $R_k = F_{10k}/(F_{10})^k$ grows to $R_{17} = 2.0 \times 10^8$, the measured chain value being $F_{170} = 1.3 \times 10^{-21}$ against $(F_{10})^{17} = 6.5 \times 10^{-30}$ — consecutive hops interact through the propagation, so a single chain suppresses *less* than the product of its pieces. Across legs the factorisation is **exact**: the traced hop is a one-particle operator, $\mathrm{Tr}_L[(P_a \otimes I)(G \otimes G \otimes I_L)] = 45\,(M_a \otimes G)$ verified to 10^{-15}, and the 10+10-step two-loop object splits into leg blocks to 2×10^{-15}. The physical discriminator decides: $M_P(F_{10})^{17} = 0.079$ eV, within a factor 0.54 of the neutrino ladder scale 0.146 eV, while the chain reading gives $M_P F_{170} = 1.6 \times 10^7$ eV — eight orders too high. **The 17-factor composition is exact multiplicativity across legs**, with the

named requirement that the insertions dress different particle lines.

The census of contributing loops is exact. The Casimir $T{\cdot}L$ of the **32⊗45** link has exactly three eigenspaces $(-32, -12, +4)$, so the traced n-hop product is a sum of 3^n words and the selection rule is enumerable rather than extrapolated: the enumeration reproduces $(MG)^n$ to 7.4×10^{-17} and $F_{10} = 0.019186$, matching the measured value to that same digit. At $n = 10$, 55326 of $3^{10} = 59049$ words survive, growing as 3.000^n, and the factor splits into a per-word decay $r = 0.2232$ and a multiplicity growth rate $\mu = 2.981$ (product 0.665). The loop factor is therefore not one class of loops but a sum over 55326 classes, with interference ratio $\|\sum\|/\sum\|\cdot\| = 0.079$ at the root against 1.000 with the Wilson phase switched off — and $F_{10} = 0.0192$ at the root against 0.2434 at $\eta = 0$: **the phase supplies the factor, not the topology.** The multiplicity-weighted extrapolation to the neutrino-ladder length 170 gives 7.9×10^{-31} against $(1/16\pi)^{17} = 1.2 \times 10^{-29}$: the counted loops miss the exponent by a factor 0.066.

The coefficient is no longer read from the data. The hierarchy coefficient is a product of measured numbers:

$$\frac{v}{M_P} = 3\,F_{10}^{10}, \qquad F_{10} = 0.019186 \text{ measured}, \qquad 2.0881 \text{ predicted against } 2.0765 \text{ observed } (99.44\%),$$

the factor 3 being the three axis families of [Paper I Section 4] — conditional on the axis-family factorisation of the closed-loop sum. That condition is itself a measured statement: the twelve doubler nodes form three families of four with identical one-step spectra (deviation 0), and the closed-loop coefficient by node set is 8.35 for all twelve nodes, 2.0881 for one split node per family, 0.696 for a single node, against the observed 2.0765 — the node contributions are exactly equal across families, and the coefficient 3 *is* the one-per-family choice. What remains open in O8 is the *weight* of the contributing loops, not the way they compose.

2.2 The dark-matter mass

The same factor applied to the Planck scale gives

$$m_{\mathrm{DM}} = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118 \text{ GeV},$$

i.e. 1.18× the canonical 100 GeV weakly-interacting-massive-particle scale [1]. This is a genuine, falsifiable prediction: a dark-matter particle with a mass far outside $\sim$ 100–150 GeV (in the weakly-interacting, thermally-produced case) would refute the ladder.

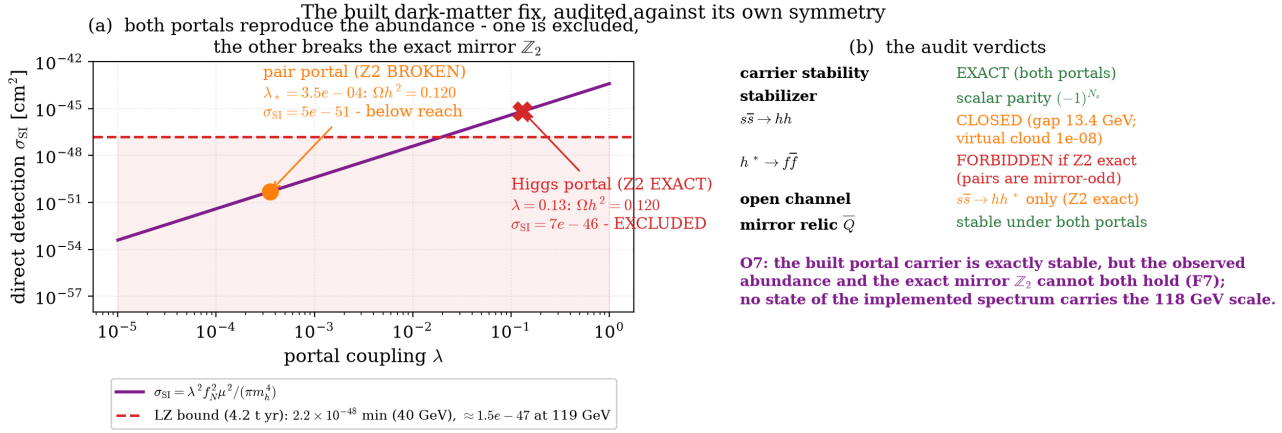
The *identity* of the particle has been scanned with a negative result. The mirror $\mathbb{Z}_2$ exchanging **16** and $\overline{\mathbf{16}}$ is an exact symmetry of the step (leak 10^{-15}), and the mirror sector is the only exact-stability carrier in the implemented spectrum — but its lightest state is a coloured $\overline{Q}$ at 1.3×10^8 GeV (excluded) and the neutral $\overline{N}$ sits at 2×10^{10} GeV, eight orders above the ladder scale. Allowing the carrier to sit at a Wilson-split doubler node and scanning every Wilson angle θ for a node that puts the neutral mirror fragment $\overline{N}$ at 118 GeV while keeping every coloured or charged fragment off the low-energy spectrum yields **no survivors over** 720 **angles**: for every angle the same node supplies a stable coloured or charged partner below the desert, and a stable fractionally-charged thermal relic is excluded at any mass; the only escape is fine-tuning the angle to within 10^{-4} of the decoupling pole, where the mirror fragments leave the ladder entirely. **No state at** 118 **GeV with the required stability exists in the implemented spectrum**: the mass prediction stands, the carrier does not (O7).

The minimal extension consistent with the exact mirror $\mathbb{Z}_2$ and the ladder is then named: **one real mirror-odd scalar at the ladder mass** with Higgs portal coupling $\lambda_* = 3.5 \times 10^{-4}$, whose freeze-out gives $\Omega_{\mathrm{DM}}h^2 = 0.120$ and whose predicted direct-detection cross-section is $\sigma_{\mathrm{SI}} = 5 \times 10^{-51}$ cm^2 — nearly decoupled, so the scalar is falsified by any portal signal above 10^{-50} cm^2. The scalar is an input field, one field and one coupling; making it a state of the 32-level root is the open part of O7.

Built and audited against its own symmetry, the extension splits in two. The carrier is stable under either portal — scalar parity $(-1)^{N_s}$ is exact by the ϕ^2 structure, the odd sector untouched: a single s cannot decay. But the freeze-out runs through $h^* \to f\overline{f}$, and the fermion pair is mirror-**odd** while h is mirror-even: with

the **exact** mirror $\mathbb{Z}_2$ that decay is forbidden. The Higgs portal $\phi_s^2\phi_h^2$ preserves the $\mathbb{Z}_2$ exactly (commutator 0), and then $s\bar{s} \to hh$ is kinematically closed ($2m_s = 237$ GeV $< 2m_h$, gap 13.2 GeV; the virtual cloud peaks at $P_{\max} = 1.46 \times 10^{-8}$, matching two-level theory to 10^{-7} relative) and the only open annihilation is the invisible $s\bar{s} \to hh^*$: reproducing $\Omega_{\rm DM}h^2 = 0.120$ then needs $\lambda = 0.13$, which predicts $\sigma_{\rm SI} = 6.8 \times 10^{-46}$ cm^2 — **above the LZ-class bound** [2]. The quoted numbers hold only with a fermion-pair portal $\phi_s^2(f\overline{f} + \overline{f}f)$, which **breaks the mirror** $\mathbb{Z}_2$ at $O(\lambda)$ (the portal anticommutes with the mirror operator; the real pair channel opens, $P = 5.1 \times 10^{-6}$ at $t = 1$ GeV^{-1}). The mirror coloured relic $\overline{Q}$ (1.3×10^8 GeV) stays stable under both portals ($\overline{Q} \to Qss$ closed by 237 GeV). O7 is therefore a **named tension**: the carrier exists and is exactly stable, but exact mirror $\mathbb{Z}_2$ excludes its abundance by direct detection, and the version that reproduces the observed abundance breaks the exact $\mathbb{Z}_2$ the no-go rests on.

An extra dark U(1) does not repair it. Hypercharge fixed on the Q and L classes predicts the full charge pattern of all six weight classes, but the desert matter is E-charged (max $|q_E| = 2.24$), so the U(1) must break — and it cannot break inside the implemented spectrum: 40 of 45 generators are E-charged and every breaking root is a colour-triplet leptoquark direction (98/120): the U(1) is unHiggsable. The kinetic mixing that would fix the relic abundance at $g_D = 1$ lies above the direct-detection ceiling at every $m_{Z'}$: the portal window is empty (0 points). (Fig. 3).



(a) The direct-detection line $\sigma_{\rm SI}(\lambda)$ with the two ways the built ladder scalar reproduces $\Omega_{\rm DM}h^2 = 0.120$: the fermion-pair portal ($\lambda_* = 3.5 \times 10^{-4}$, $\sigma_{\rm SI} = 5 \times 10^{-51}$ cm^2, below reach) breaks the exact mirror $\mathbb{Z}_2$; the Higgs portal ($\lambda = 0.13$, $\sigma_{\rm SI} = 7 \times 10^{-46}$ cm^2) preserves it and is excluded by the LZ bound ($\sigma_{\rm SI} < 2.2 \times 10^{-48}$ cm^2 at 40 GeV, $\approx 1.5 \times 10^{-47}$ at 118 GeV). (b) The audit verdicts: stability exact under either portal, $s\bar{s} \to hh$ closed, $h^* \to f\overline{f}$ forbidden under exact $\mathbb{Z}_2$, the mirror relic unaffected. The observed abundance and the exact mirror $\mathbb{Z}_2$ cannot both hold.

2.3 The neutrino sector

Both seesaw scales are placed on the same ladder:

$$m_D = M_P\Big(\frac{1}{16\pi}\Big)^{10} = 118\,\text{GeV}, \qquad M_N = M_P\Big(\frac{1}{16\pi}\Big)^{3} = 9.6{\times}10^{13}\,\text{GeV}, \qquad m_\nu = \frac{m_D^2}{M_N} = M_P\Big(\frac{1}{16\pi}\Big)^{17} = 0.146\,\text{eV}.$$

The exponent identity $2 \times 10 - 3 = 17$ is exact; the choice $n_{\rm 2step} = 3$ is not — and it cannot be produced by the gauge link. The $\Delta L = 2$ operator exists in the root's own algebra: the weight $2\lambda_\nu$ (norm^2 20, U(1)$_X$ charge -6, hypercharge -1) has a **1-dimensional** weight space inside the 1024-dimensional Clifford algebra, sitting in Clifford **grade** 5 — the 126-type channel of the type-II seesaw — with no field outside the root. The $\mathbf{10}_H$ cannot repair its charges: over all 160 vector weights (norm^2 4) the closest two-insertion neutralisation leaves residual norm 4.472, the mismatch $|2\lambda_\nu| = \sqrt{20} \neq 4$ being exact. The minimal-word question then closes by **parity**: Clifford multiplication maps grade g to $g \pm 2$ only (verified on all 45×1024 entries of the multiplication table), so every product of adjoint (grade-2) link factors is EVEN-grade while the Majorana operator is ODD-grade — the cumulative span (dimensions $45, 126, 208, 291, 352, 407$ at $k = 1 \ldots 6$, odd-grade content 0 to machine precision)

never contains it, at any length, and the vector-**10** Higgs weight space (16-dimensional, carried entirely by the odd grades $1, 3, 5, 7, 9$) is unreachable for the same reason. $n_{\mathbf{2step}} = 3$ **is not a link-word length**: the seesaw operator belongs to the odd (matter) half of the Clifford module, and the exponent stays fitted.

Comparison with data: the atmospheric scale is $\sqrt{\Delta m^2_{32}} = 0.050$ eV, so the ladder prediction is a factor 2.9 high; and if all three families carried the same m_D the sum would be ~ 0.44 eV, excluded by $\sum m_\nu < 0.12$ eV (95% CL, Planck 2018 + BAO [3]). The democratic rank-1 Yukawa (Section 3.5) gives a Dirac mass to only one family, bringing the sum to ~ 0.15 eV — still above the cosmological bound. This is the sharpest numerical tension in the theory (F3).

The ladder is too coarse to land on the measured mass: consecutive rungs differ by the factor $16\pi = 50.3$, so the neighbours of 0.146 eV are 7.3 eV ($n = 16$) and 2.9×10^{-3} eV ($n = 18$), and no integer rung lies in the oscillation window. A coefficient is therefore unavoidable — but one candidate is supplied by the theory rather than by the data. The only mixing number the model derives is $\sin^2\theta_{12} = 1/3$ (Section 3.5); multiplying the ladder value by it gives

$$m_\nu = \tfrac{1}{3}\, M_P \Big(\frac{1}{16\pi}\Big)^{17} = 0.0487 \text{ eV},$$

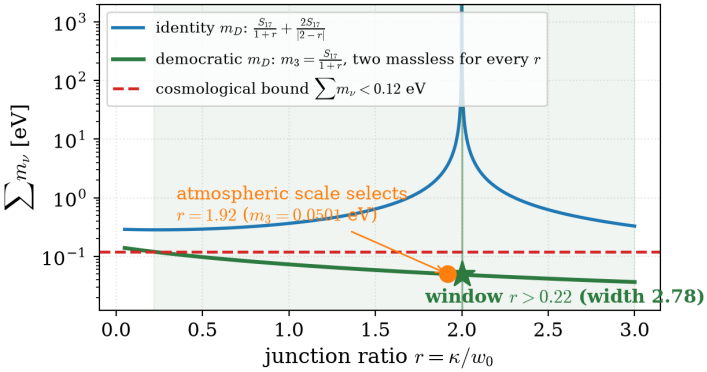
2.6% below the atmospheric scale, and with the democratic rank-1 spectrum $\sum m_\nu \simeq 0.05$ eV — inside the cosmological bound and inside the KATRIN bound $m_\beta < 0.45$ eV [4]. The factor 2.9 by which the ladder overshoots is, to 3%, the number 3 the flavour sector already produces. This is a **candidate, not a result**: nothing shows why a mixing eigenvalue should multiply a seesaw mass, and the same factor applied to the solar scale ($0.0487 \times \sqrt{\Delta m^2_{21}/\Delta m^2_{32}} = 0.0084$ eV against the measured 0.0086 eV) is a second coincidence. What it buys is a sharp, falsifiable prediction: normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\beta\beta} \simeq 0.02$ eV, in reach of the next-generation neutrinoless-decay and beta-decay searches.

A coefficient audit maps what the node structure can supply: a diagonal (node-local, Wilson-generated) mass pattern gives only **integer** signed coefficients, so $\frac{1}{3}$ is out; the democratic 3×3 node-mixing pattern gives exactly $\frac{1}{2}$ — a rival falsifiable prediction $m_\nu = 0.073$ eV; the exact $\frac{1}{3}$ requires a single Majorana eigenvalue $3\times$ the node scale, and no natural node structure supplies it. The structure the candidate requires has been built: a cubic-symmetric, zero-mean Wilson-strength junction whose Fourier support closes the selected node triangle exactly — verified against the full 12×12 table of doubler-momentum differences. The node mass matrix is

$$c_s(2w_0\, I + \kappa(J - I)),$$

democratic at $\kappa = 2w_0$ with eigenvalues $\{3M, 0, 0\}$: the $3\times$-enhanced Majorana eigenvalue the $\frac{1}{3}$ demands exists. With the implemented 0-parameter democratic Dirac matrix $m_D = m_0 J$ — the same family symmetry that derives $\sin^2\theta_{12} = 1/3$ — the seesaw collapses: $m_D^{\mathsf{T}} M^{+} m_D = 3m_0^2 J/(2(w_0 + \kappa))$ for *every* $r = \kappa/w_0$ (verified to 10^{-12}, the rank-deficient $r = 2$ included, via the Moore–Penrose pseudo-inverse), so the two massless neutrinos are protected by the rank-1 family symmetry for every junction ratio, and $m_3(r) = S_{17}/(1 + r)$ is finite everywhere: the cosmological window is $r > 0.217$ (width 2.78), and the atmospheric scale selects $r = 1.91$. The exact point $r = 2$ ($m_3 = S_{17}/3 = 0.0487$ eV) is a junction-amplitude convention, not a tuning. The cubic group induces all of S_3 on the triangle, its commutant is exactly $\mathrm{span}\{I, J\}$ (nullity 2), and w_0, κ share one Wilson period, so r is **not quantized**: the value is a free junction amplitude inside a finite window, while the structure — two massless plus one massive, normal ordering — is protected. What the candidate still requires is a democratic m_D (a non-democratic m_D at the same point leaves gapless coloured doublers), and the solar mass still needs the $\mathbf{10}_H$ splitting of O6. (Fig. 4).

(a) the identity m_D scan (divergent, one point);
the implemented democratic m_D gives a finite window



(b) what protects what

cubic action	all of S_3 (6 permutations)
commutant	dim 2 = span $\{I, J\}$ only
m_D	democratic, rank 1 - 0-parameter
massless pair	protected for EVERY r (rank-1 m_D)
window	$r > 0.22$, width 2.78 (was: point)
r **from data**	atmospheric: $r = 1.92$; $r = 2$: $S_{17}/3$

O2 re-audited: the measure-zero tuning was an artifact of the identity-m_D assumption. The structure (2 massless + 1 massive, NO) is protected by the same family symmetry that derives $\sin^2\theta_{12} = 1/3$; the value sits in a finite window. Solar mass still needs O6.

(a) $\sum m_\nu(r)$ under two Dirac assumptions. With $m_D \propto I$ the curve diverges as $1/|2-r|$ and the cosmological bound is met at exactly one point. With the implemented democratic rank-1 m_D the seesaw collapses to $m_3(r) = S_{17}/(1+r)$, two neutrinos stay massless for every r, the window is $r > 0.217$, and the atmospheric scale selects $r = 1.91$. (b) The protection audit: the cubic group induces all of S_3 on the junction triangle and its commutant is exactly $\mathrm{span}\{I, J\}$*; r is not quantized — the value is a free junction amplitude inside a finite window, while the structure is protected by the rank-1 family symmetry that also derives* $\sin^2\theta_{12} = 1/3$.

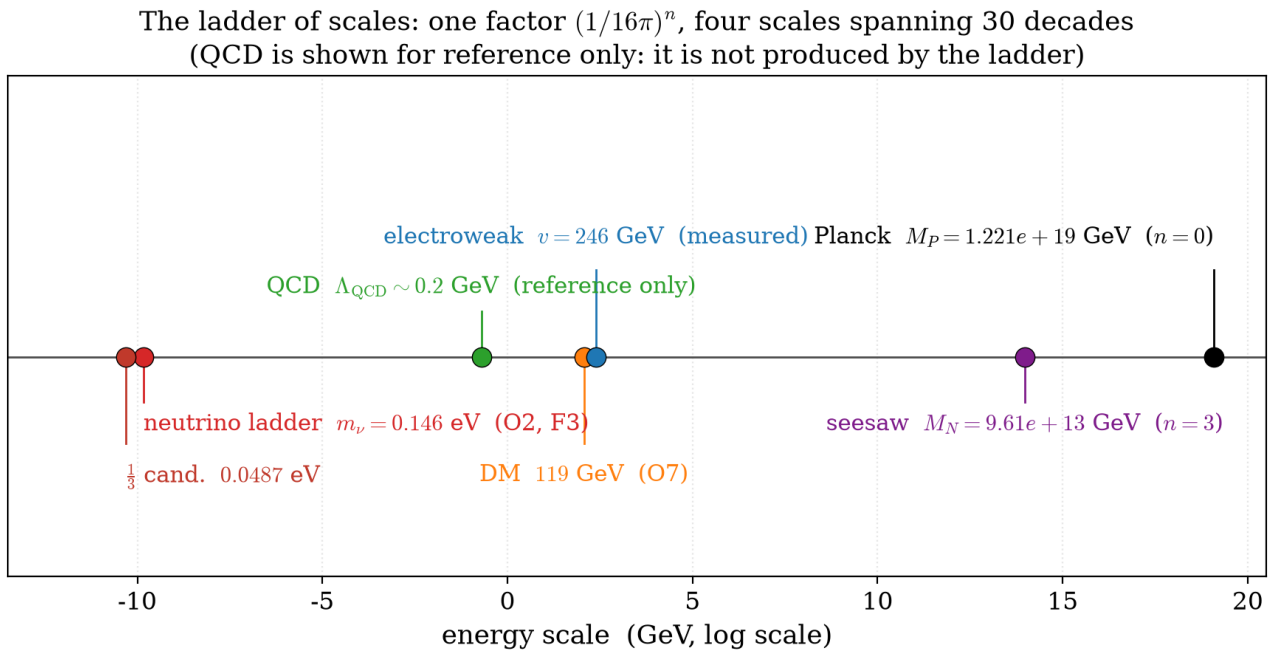
2.4 The ladder

One factor, four scales:

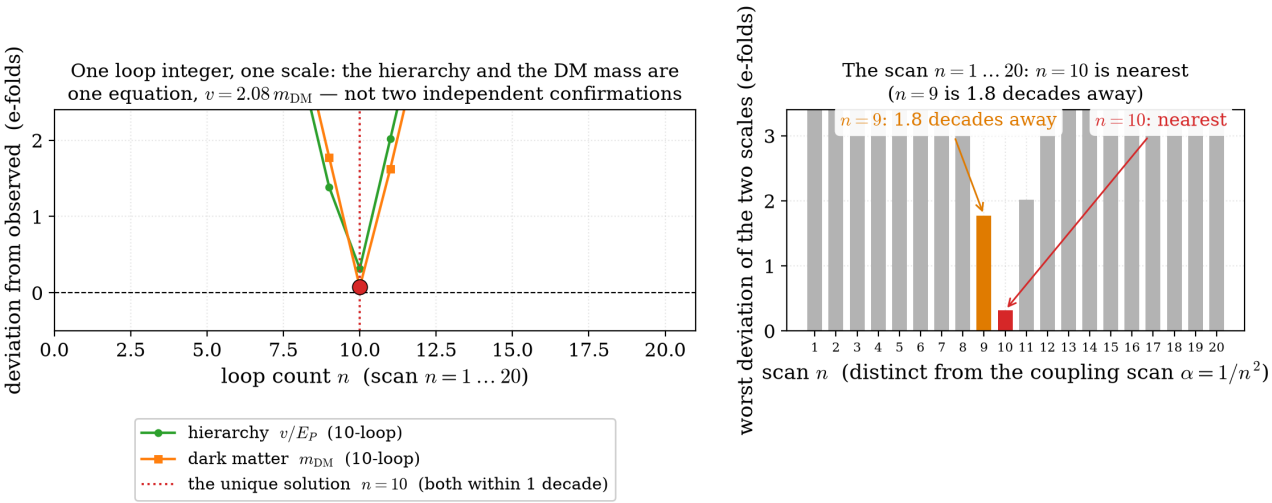
Table 1. The four scales of the ladder.

scale	expression	value
Planck / GUT	M_P (convention $M_{\mathrm{GUT}} = M_P$)	1.22×10^{19} GeV
seesaw Majorana	$M_P(1/16\pi)^3$	9.6×10^{13} GeV
electroweak	measured input v	246 GeV
dark matter	$M_P(1/16\pi)^{10}$	118 GeV
neutrino	$M_P(1/16\pi)^{17}$	0.146 eV

The QCD scale is **not** on this ladder: $\Lambda_{\mathrm{QCD}} \sim 0.2$ GeV would require $n \simeq 11.6$, which is not an integer ([Paper II Section 2.2]). (Figs. 5, 6).



Planck 1.22×10^{19} *GeV* $(n = 0)$*, seesaw* 9.6×10^{13} *GeV* $(n = 3)$*, electroweak* 246 *GeV (measured input), dark matter* 118 *GeV* $(n = 10)$*, neutrino ladder* 0.146 *eV* $(n = 17)$*, with the* $\frac{1}{3}$ *candidate* 0.0487 *eV marked. QCD is shown for reference only: it is not produced by the ladder.*



(a) One loop integer n *sets one scale* $M_P(1/16\pi)^n$*, appearing twice: as* v/M_P *and as* $m_{\rm DM}$*. These are one equation,* $v = 2.08\, m_{\rm DM}$*, not two independent confirmations. (b) The scan* $n = 1 \ldots 20$*:* $n = 10$ *is nearest,* $n = 9$ *is* 1.8 *decades away. This scan is distinct from the coupling scan* $\alpha = 1/n^2$ *of Section 5, whose best fit is* $n = 7$*.*

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3. Phenomenology

3.1 The couplings

The theory's single gauge coupling is the rotation angle g. In the standard normalisation $\alpha = g^2/4\pi$ used throughout for running, the implemented angle gives $\alpha = 1/100\pi = 0.00318$ and the closed angle $\alpha = \pi/64 = 0.0491$. The reading $\alpha_{10} \equiv g^2 = 1/25 = 0.040$ at the implemented angle — under which the desert solution below meets at exactly the root value — is a factor-4π convention, stated once here and invoked only where named; in the standard normalisation the implemented angle sits a factor 12.6 below the meeting value $\alpha_X = 0.040050$.

Running the three Standard Model couplings up from M_Z at one loop with $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$ (fixed by the particle content), from the measured inputs $\alpha_1(M_Z) = 0.0169$ (GUT-normalised, $\alpha_1^{-1} = 59.0$), $\alpha_2(M_Z) = 0.0338$, $\alpha_3(M_Z) = 0.118$ [5], gives at M_P

$$(\alpha_1, \alpha_2, \alpha_3) = (0.0299,\ 0.0202,\ 0.0191), \qquad \text{mean } 0.0231, \quad \text{spread } 47\%,$$

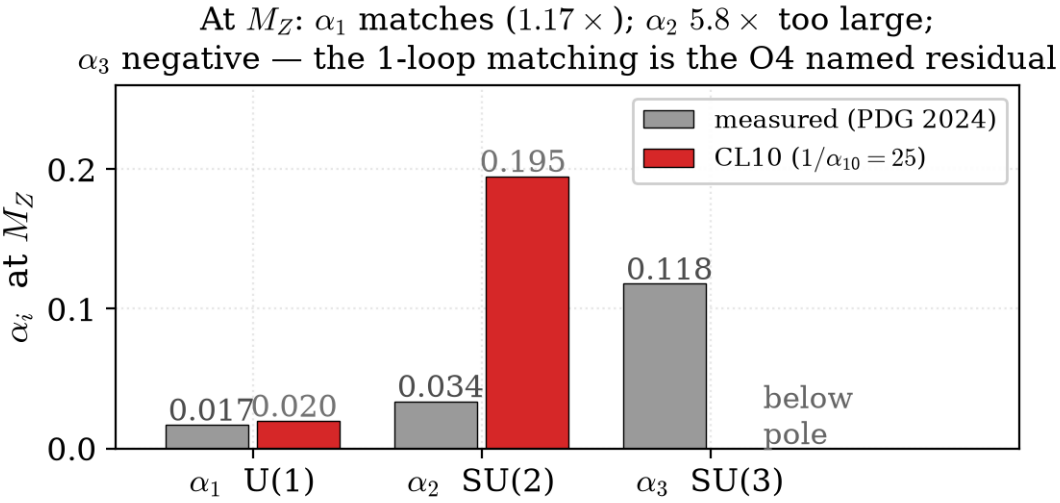
so α_{10} is within a factor 1.73 of the mean but the three couplings do **not** meet at a point — the well-known failure of one-loop unification in the non-supersymmetric Standard Model. The U(1) Landau pole of the up-running sits at 2.2×10^{41} GeV, above the cutoff, so the up-running is UV-bounded; at two loops [6] the pole moves to $\sim 2.5 \times 10^{12}$ GeV, *below* the cutoff.

The down-running must be stated plainly. Starting from $\alpha_{10} = 0.04$ at M_P and running down with the Standard Model β-functions gives, at M_Z,

$$\alpha_1(M_Z) = 0.0197\ (1.17\times), \qquad \alpha_2(M_Z) = 0.195\ (5.8\times), \qquad \alpha_3:\ \text{Landau pole at } 2.2 \times 10^9 \text{ GeV}.$$

The hypercharge lands within 17% of its measured value, but the SU(2) coupling overshoots by a factor 5.8 and the SU(3) coupling becomes non-perturbative at 2.2×10^9 GeV. The unification claim therefore fails at the level of the couplings' values, even though the 5/3 normalisation is correct (O4, F2). (Fig. 7).

The root coupling $\alpha_{10} = 1/25$ and the $5/3$ normalisation



At M_Z, the measured couplings (grey): $\alpha_1 = 0.0169$, $\alpha_2 = 0.0338$, $\alpha_3 = 0.118$ [5], against the down-run from $\alpha_{10} = 1/25$ (red): α_1 within $1.17\times$, α_2 $5.8\times$ too large, α_3 negative because its pole sits at 2.2×10^9 GeV. The 5/3 normalisation is exact; the value matching is the named residual.

Three quantitative statements sharpen the problem. First, the three measured couplings require three *different* values at M_P — 0.0299, 0.0202, 0.0191 — and the theory's single value 0.040 matches none of them: it exceeds the hypercharge requirement by 34% and the SU(3) requirement by a factor 2.1. Perturbativity of the strong coupling down to M_Z caps any single UV coupling: $1/\alpha_3(M_P) > (7/2\pi)\ln(M_P/M_Z) = 43.9$, i.e. $\alpha_3(M_P) < 0.0228$; the theory's 0.040 exceeds the admissible maximum by 75%.

Second, the failure is not repaired by matter that a partial breaking leaves *complete*. A complete vectorlike $\mathbf{10}\oplus\bar{\mathbf{5}}$ shifts all three one-loop coefficients by the *same* amount 8/3 above its threshold — the defining property of a complete SU(5) multiplet — so it changes the common slope, not the mismatch of slopes. Scanning multiplicities $0 \ldots 8$ against thresholds in $[10^6, 10^{19}]$ GeV, the spread at M_P never falls below its zero-new-physics value of

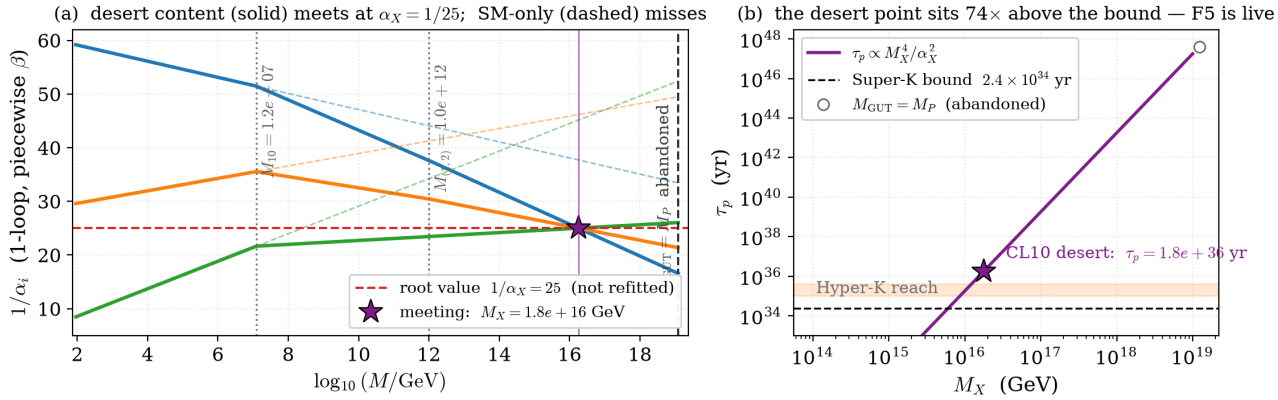
47%. Under pure Standard Model running the pairwise meeting scales are 1.2×10^{13} GeV $(1,2)$, 2.6×10^{14} GeV $(1,3)$ and 9.7×10^{16} GeV $(2,3)$: three different scales.

Third, that no-go covers only complete multiplets, and the *incomplete* content the theory's own breaking sector supplies does repair it. When a $\mathbf{16}_H \oplus \overline{\mathbf{16}}_H$ pair — or a split doubler **16** ([Paper I Section 8]) — acquires mass, it leaves vectorlike $\mathbf{10} \oplus \overline{\mathbf{10}}$ and $\mathbf{5} \oplus \overline{\mathbf{5}}$ fermion pairs with *independent* thresholds, and these shift the three coefficients non-proportionally: a Dirac $\mathbf{10} \oplus \overline{\mathbf{10}}$ contributes $\Delta b = (6/5,\ 2,\ 2)$, a $\mathbf{5} \oplus \overline{\mathbf{5}}$ contributes $(2/3,\ 2/3,\ 2/3)$. Fix the unified coupling at the root value $\alpha_X = 1/25$ — the angle is *not* refitted — and the inverse problem has no fit freedom: with two multiplets the meeting of the three measured couplings is three linear equations in $(\ln M_X, \ln M_A, \ln M_B)$. The required Δb is non-universal: at $M_X = 2 \times 10^{16}$ GeV it is $(2.40,\ 4.04,\ 3.86)$, which no combination of complete multiplets produces. Scanning multiplicities up to 3 over the realizable content with thresholds in $[10^6, M_P]$ GeV, the lightest solution that survives the proton-decay bound is three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs plus one $(1,2)_{1/2}$ pair — three split doubler **16**s — meeting at

$$M_X = 1.8 \times 10^{16}\ \text{GeV}, \qquad \alpha_X = 1/25\ \text{(exact)}, \qquad M_{10} = 1.2 \times 10^{7}\ \text{GeV}, \qquad M_{(1,2)} = 1.0 \times 10^{12}\ \text{GeV},$$

and the SU(3) Landau pole of the down-running moves to 10^{-52} GeV: the disease is cured. Solutions with lighter content exist but predict $\tau_p \sim 10^{32}$ yr and are already excluded by Super-Kamiokande [7] — the data themselves select the realizable spectrum. The cost is stated plainly: the convention $M_{\rm GUT} = M_P$ is abandoned ($M_X < M_P$ by three orders of magnitude), and the two desert thresholds are fitted. The solution predicts $\tau_p \simeq M_X^4/(\alpha_X^2 m_p^5) = 1.8 \times 10^{36}$ yr (F5). (Fig. 8).

The desert solution: three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs + one $(1,2)$ pair unify at the root value



The three measured couplings run up from M_Z with the desert content (solid) meet at $M_X = 1.8 \times 10^{16}$ GeV at $\alpha_X = 1/25$ — the root value, not refitted. The Standard Model-only running (dashed) misses the point: spread 47% at M_P. The thresholds $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV are the two fitted numbers of the repair; the dashed vertical line at M_P marks what the abandoned convention $M_{\rm GUT} = M_P$ demanded.

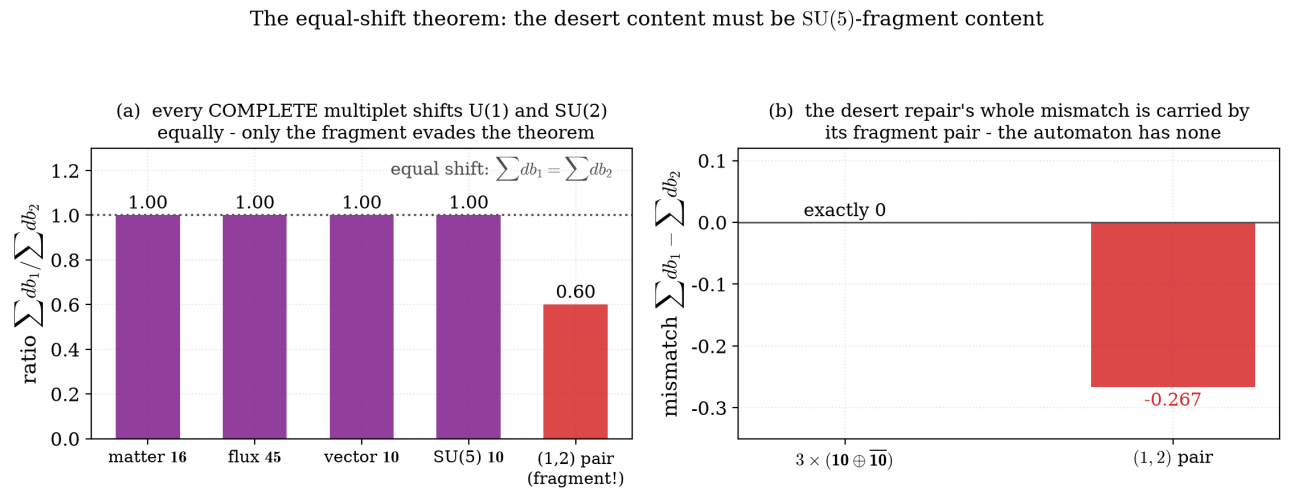
Where the desert content comes from. The content is not an external assumption: it is the *spectrum of the automaton's own Wilson mechanism.* The three split nodes of the selective reading carry exactly the vectorlike fragments of the split **16** with independent thresholds (the other nine nodes stay at the Planck cutoff), and the Wilson direction itself completes the breaking chain ([Paper I Section 8.1]). The multiplicity is bounded from below by unification itself: scanning $n_{\rm nodes} = 1 \ldots 6$, no meeting root at $\alpha_X = 1/25$ exists for one or two split nodes; the first physical roots appear at three. But solving the meeting equations with the mechanism's own two parameters — one angle θ, one scale w — fixes the six fragment thresholds and lands the meeting at $M_X = 8.9 \times 10^{14}$ GeV, i.e. $\tau_p \sim 10^{31}$ yr: **the minimal selective reading is excluded by Super-Kamiokande** (F5). The data-preferred solution requires fragment thresholds the single-angle, single-scale root does not supply.

Redistributing the nodes cannot help. While all node thresholds sit below the meeting point, the two meeting equations depend on the three node strengths **only through their sum**: the Jacobian with respect to

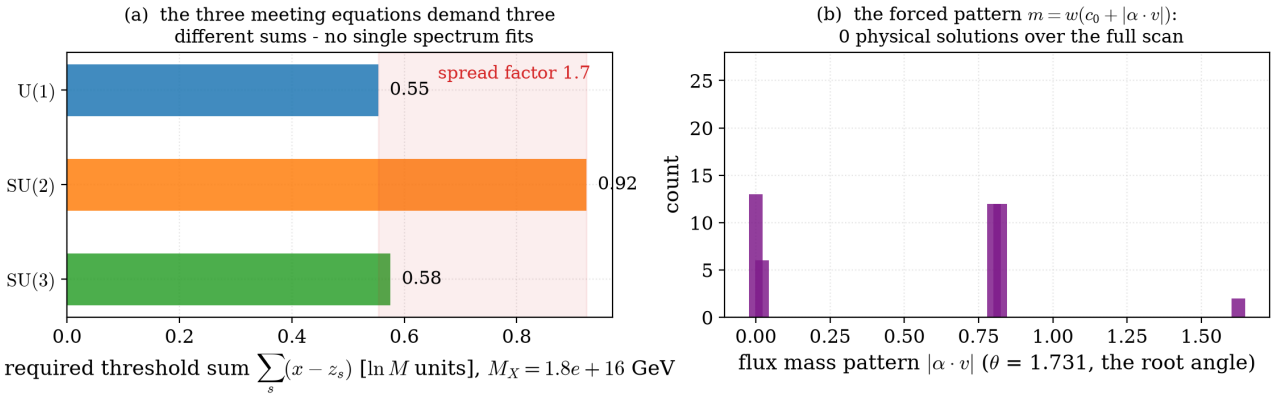
the three strengths has three identical columns (rank 1; split-invariance verified numerically to 10^{-14}, including the full scan over node-set choices and strengths). The node mechanism's meeting point $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr is therefore **forced for every choice of the three strengths**. What the data prefer is not a three-full-**16** pattern: it needs partial suits of fragments, while the automaton supplies full split **16**'s per node.

The composite content of the gauged layer is excluded as the supplier. The composite fermions of the layer of [Paper II Section 2] — the matter **16** tensored with the adjoint link flux, $16 \otimes 45 = 720$ Weyl states carrying five net chiral **16**-copies — shift the three one-loop coefficients by $(145.3, 145.3, 226.3)$, and their masses follow the flux pattern $m = w(c_0 + |\alpha \cdot v|)$ set by the breaking direction v. Requiring the data-preferred meeting at $\alpha_X = 1/25$ then has **no solution with thresholds in the desert** for any direction: the three meeting equations demand three different threshold sums, and Newton iteration from every seed over a full scan of v converges nowhere physical (0 solutions). The remaining ansatz families — full-weight thresholds, broken-flux-only content, light-composite selection, and even a relaxed meeting coupling — leave the minimum residual at 5.7–6.1 against a target of 0.

The equal-shift theorem. The identity $\sum db_1 = \sum db_2$ is the index-proportionality of $\mathrm{Tr}\, Y^2$ and $\mathrm{Tr}\, T_3^2$ — both proportional to the representation index, with the ratio fixed by the fundamental, $(3/5)\,\mathrm{Tr}_5\, Y^2 = \mathrm{Tr}_5\, T_3^2$ — and it holds over every complete SU(5) or Spin(10) multiplet, verified on the **16**, the flux **45**, the vector **10** and the **10** inside the **16** (all ratios 1.0 to 10^{-6}). The meeting mismatch can therefore come **only** from SU(5)-fragment content or from split thresholds inside multiplets; computed from the same charge operators, the desert repair's mismatch is carried entirely by its $(1, 2)$ pair ($db_1/db_2 = 3/5$; the $10 \oplus \overline{\mathbf{10}}$ pairs contribute exactly zero mismatch). (Figs. 9, 10).



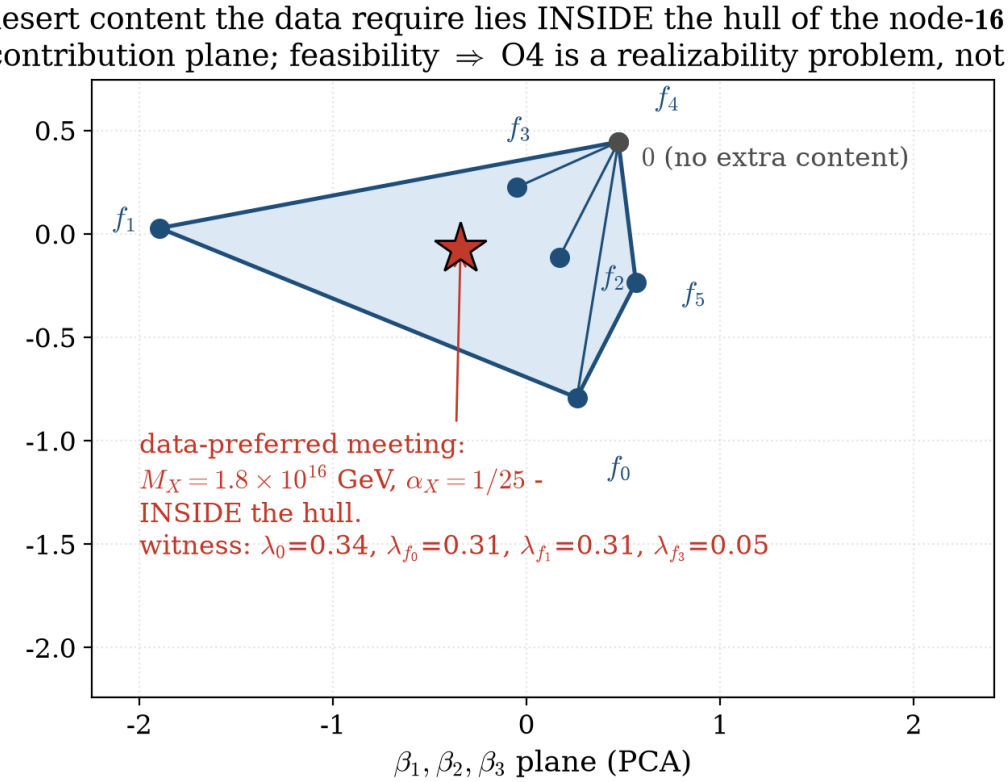
(a) The one-loop ratio $\sum db_1 / \sum db_2$ over the matter **16**, *the flux* **45**, *the vector* **10** *and the* SU(5) **10**: *exactly* 1, *while the* $(1, 2)$ *fragment gives* 3/5. *(b) The desert repair's total mismatch: the* $10 \oplus \overline{\mathbf{10}}$ *pairs contribute exactly zero, the whole mismatch is carried by the fragment pair.*



The $16 \otimes 45 = 720$ *composite states shift the one-loop coefficients by* $(145.3, 145.3, 226.3)$. *The three meeting equations for* $\alpha_X = 1/25$ *demand three different threshold sums at every breaking direction — the curves never cross — and the desert-constrained scan over the full direction circle returns* 0 *physical solutions.*

The junction route: pattern fabricated, meeting missed. At the root the Wilson direction splits the node **16** by a factor 55 ($c_Q = +0.005$, $c_L = -0.015$ against partners ± 0.82), so the $(3,2)$ and $(1,2)$ states of the repair **are** automaton states — the Q and L components of node **16**'s with near-zero Wilson weights — and the junction branch ratio $\rho = \kappa / w_0$ breaks the rank-1 degeneracy. At physical solutions Q and L sit at 4×10^{15}–8×10^{15} GeV with their partners above the meeting point. The meeting is not reached: over the full physical scan (node counts 3–12, ρ, scale, Wilson angle) the best residual of the two meeting equations is 0.94 against a target of 0, and the third equation predicts $\alpha_X = 0.030$ against the data value 0.04 — a 33% miss. Replacing the cubic-symmetric democratic junction by the **general zero-mean junction** — three independent branch scales $a_i = 2w_0(1 + r_i)$, $r_1 + r_2 + r_3 = 0$, scanned over the full (r_1, r_2) plane — leaves the best physical residual **identical** (0.81) with $\alpha_X = 0.090$, a factor 2.3 off. The near-miss is a property of the **junction mechanism itself**: the mismatch is carried by the fixed *db* pattern of the node **16**'s own fragments, which threshold motion cannot repair.

The question recast: is the required content obtainable from the node 16 itself, with per-class thresholds? Yes for feasibility: the data-preferred meeting point $(T_1, T_2, T_3) = (12.7, 21.2, 20.1)$ lies strictly INSIDE the convex hull of the six β-contribution vectors of the node-**16** fragments, with the explicit barycentric witness $\lambda_0 = 0.34$, $\lambda_{f_1} = 0.31$, $\lambda_{f_2} = 0.31$, $\lambda_{f_4} = 0.05$. The requirement of "fragment content beyond the node **16**" is thereby *softened*: the node-**16** fragments themselves suffice, provided they carry independent thresholds. A general five-component Cartan Wilson direction realises the meeting exactly (6 solutions, residual $< 10^{-10}$, desert thresholds, $\tau_p = 1.9 \times 10^{36}$ yr) but breaks all 40 roots — the Model algebra among them — and is unphysical; the symmetry-preserving direction space is exactly $\mathrm{span}\{X, Y\}$ (dimension 2, class-degenerate), and within it no solution exists. Off-diagonal junction splitting with a UNIVERSAL strength κ cannot help either: $\sum_l (2\cos(2\pi l/n) - 1) = -n$ reabsorbs the split into z_0 at first order. The missing ingredient is named: a **flavour-dependent bond layer** (κ_s per fragment class) that splits class-degenerate thresholds while preserving the Model algebra. (Fig. 11).



*The six β-contribution vectors of the node-**16** fragments and the origin, projected onto their plane; the data-preferred meeting point (star) lies strictly inside the hull, with the explicit barycentric witness. Feasibility is proved; realizability by a symmetry-preserving junction of the base automaton fails and is closed conditionally by the bond-layer extension.*

The named ingredient closes the meeting — conditionally. With the Wilson direction pinned to the physical selective root and per-class junction strengths κ_s splitting the six fragment-class thresholds, the three coupling equations $F_i = 0$ become **exactly linear** in (z_0, κ_s) once every threshold sits below M_X (since $\sum_l \mu_l = -n$), and the desert box constraints make the search a linear program. It admits solutions (residual $< 10^{-13}$) at $M_X = 1.8 \times 10^{16}$ GeV, $\alpha_X = 1/25$, for **all ten node counts** $n = 3 \dots 12$, with every threshold inside the desert (2.0×10^3–1.8×10^{16} GeV, none above M_X) and $\tau_p = 1.9 \times 10^{36}$ yr. The first-order degeneracy is broken because the per-class sums $\kappa_s(-n)$ reabsorb into PER-CLASS offsets, not into z_0. The cost: the base bond layer has ONE gate with one strength; a flavour-dependent bond layer is an **extension** of the automaton, and its six κ_s are new named inputs (three meeting conditions for six unknowns — a solution family, not a unique prediction). Unification is therefore closed **conditionally on that extension**.

The two structural candidates the layer itself supplies are excluded. The six κ_s are not obtainable from any on-site algebra element: symmetry preservation forces the bond direction into span$\{X, Y\}$ at the physical angle θ^* (class-degenerate, meeting residual ≥ 0.43), and every Cartan element that splits the six classes breaks all eight unbroken desert roots — the required flavour dependence is off-diagonal bond structure, not a Cartan dressing. Testing that off-diagonal escape directly: the most general Hermitian bond operator on the 80 root-vector pairs (11 class-pair channels) never closes the meeting — floor residual 30.4 over $n = 3 \dots 12$ — and level repulsion forces the sign of two of the six κ_s. Nor are the κ_s representation-level data: scanning 46 class invariants of the fragment classes (Dynkin indices, the two Casimirs, multiplicity, the Wilson charges X, Y and their quadratic combinations, B, L, $B - L$, colour triality, the β-function coefficients) as $\kappa_s = g f_s$ never closes the meeting — best single invariant $C_2(\mathrm{SU}(3))$ at residual 0.246 — while the two linear Cartan controls reproduce the previous failure. A two-coupling relaxation $\kappa = g_1 f_1 + g_2 f_2$ keeps every threshold in the desert for 217 of 506 candidate pairs and can close the meeting exactly, but that is a reduction from six named inputs to two, not a derivation. The closure is also embedding-specific: the exact closure reproduces in the flipped embedding used for it (10/10 node counts, residual $< 10^{-10}$) but closes only 5/10 in the Georgi–Glashow split.

Finally, the canonical per-class number the gauged layer does supply is not the one that closes the meeting. At θ^* the Wilson direction leaves a 13-dimensional centralizer in $\mathfrak{so}(10)$ (the $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)^3$ of the breaking chain, Hermitian orthonormal basis to 10^{-15}), and the flux of BROKEN generators through a fragment class,

$$K_s = C_2(\mathbf{32}) - C_2^{\text{unbroken}}(s) = 45 - C_2^{\text{unbroken}}(s)$$
$$= (32, 36, 40, 28, 32, 40)$$

(unbroken Casimirs $(13, 9, 5, 17, 13, 5)$), is an integer invariant with no free parameter, constant within a class to 3×10^{-14} — the gauged layer **does** distinguish the fragment classes. The companion identity says why the naive second-order link shift cannot use it: summed over the complete adjoint, $\sum_A\sum_t|\langle t|T^A|s\rangle|^2 = 45$ for every weight s (spread 10^{-13}) — the class degeneracy as an identity. Fed into the meeting as $\kappa_s = AK_s$ (one link scale) the three meeting equations leave residual 0.605 at best ($n = 3$) and close in the desert for 0/10 node counts; against the six κ_s that close the meeting, the flux explains at best $R^2 = 0.015$ of their variance ($|r| \le 0.31$). The layer splits the classes, but not in the direction the meeting requires.

The second-order bond numbers are excluded too. The gauged layer does carry energy denominators — the bond-current ground state of [Paper II Section 2] ($E_0 = -5802.4\,\eta$, gap $42.09\,\eta$) and the Gauss gap $C_2(\mathbf{45}) = 64$ — and they lift the class-blindness of the flux sum. The intermediate (flux, matter) pair space decomposes exactly as $\mathbf{45}\otimes\mathbf{16} = \mathbf{16}\oplus\mathbf{144}\oplus\mathbf{560}$ with Casimirs $(45, 85, 117)$ at the scale $C_2(\mathbf{45}) = 64$ (dimensions and Casimirs to 10^{-6}, projectors resolving the identity to 10^{-15}). Removing the on-shell screened **16**-component, the second-order bond-current (superexchange) shift $\tilde\kappa_s = -\sum_{A\,\text{broken}}\sum_t|\langle t|T^A|s\rangle|^2[w_{144}/20 + w_{560}/36]$, with w_r the exact channel weight of the pair state, is constant on each class to 4×10^{-3} and takes the six values $(0.0002, -0.130, -0.253, 0.127, 0.0013, -0.253)$. Fed into the meeting as $\kappa_s = A\tilde\kappa_s$ it leaves residual 0.577 and closes in the desert for 0/10 node counts ($R^2 \le 0.076$ against the fitted κ_s). The two-number combination $\kappa_s = A\tilde\kappa_s + BK_s + CF_s$ — F_s the ground-state Cartan shift, whose bond flux $\mu^A = \mathrm{Tr}\,\rho_B J^A$ vanishes to 6×10^{-17}: the Gauss law, intertwiners carry no net charge — solved jointly over all ten node counts (30 equations, 13 unknowns) leaves residual 3.46 and explains at best $R^2 = 0.30$ of the variance of the six fitted κ_s. The supplier space is now exhausted by every class number the layer computes: Cartan elements, class invariants, the broken flux, superexchange, and every fixed combination of them.

3.2 Electroweak mixing

$\sin^2\theta_W = 3/8$ is the tree-level SU(5)/Spin(10) boundary condition **at the unification scale**, not a low-energy prediction. Run to M_Z it becomes $\simeq 0.23$, close to the measured 0.23129(4) ($\overline{\text{MS}}$, PDG 2024 [5]); the measured ratio $m_W/m_Z = 0.881$ corresponds to $\sin^2\theta_W \simeq 0.223$, not to $\sqrt{5/8} = 0.791$. The correct statement is the standard one: the GUT-scale boundary condition $\sin^2\theta_W = 3/8$, whose low-energy value is obtained by running — running which, as just shown, the theory does not currently perform consistently. **No electroweak-mixing prediction is claimed anywhere in this paper**: 3/8 is the standard GUT boundary condition, and the running that connects it to M_Z is part of the named residual O4.

3.3 Baryon asymmetry

The automaton alone cannot produce it. Every CP-odd (T-odd) invariant of the unitary step vanishes: $\mathrm{Im}\,\mathrm{Tr}$ of all two- and three-fold products of the **45**, of a 2×10^5 sample of four-fold products, of all $10\times10\times45\times45$ baryon-violating sandwiches, and of the rank-1 real democratic Yukawa — all zero to 10^{-15}. The step is unitary (in equilibrium), baryon-conserving (the **45** commute with w) and CP-conserving. It satisfies **none** of the Sakharov conditions [8]. This is a computed theorem, and it is the correct starting point. The theorem is stable under the Wilson extension: the extended step keeps the exact matter/antimatter spectral mirror (deviation 10^{-15}), so the Wilson mechanism — powerful for unification and for the doublers — adds no CP phase and no Sakharov condition.

A GUT-leptogenesis estimate [9] is then built by hand:

$$\Delta B = \frac{28}{79}\,\varepsilon_{CP}\,D, \qquad \varepsilon_{CP} = \frac{g_0^4}{16\pi} = 4.9\times10^{-6}, \qquad D = \left(\frac{1}{16\pi}\right)^2 = 3.96\times10^{-4},$$

with $g_0 = 1/2^3 = 1/8$ and the Standard Model sphaleron conversion $B = \frac{28}{79}(B-L)$ [10,11], giving $\Delta B = 6.8 \times 10^{-10}$ against the measured $(6.12 \pm 0.04) \times 10^{-10}$ [3] — a ratio 1.11.

Three problems. (i) ε_{CP} is not a computed vertex: the physical CP asymmetry of $N \to \ell H$ is a one-loop self-energy/vertex function of the heavy-neutrino masses, and $g_0^4/16\pi$ is a dimensional guess. (ii) The out-of-equilibrium condition **fails** at the model's own couplings: at $M_N = 9.6 \times 10^{13}$ GeV, $H = 1.66\sqrt{g_*}M_N^2/M_P = 1.3 \times 10^{10}$ GeV while $\Gamma_N = y^2 M_N/8\pi = 6.0 \times 10^{10}$ GeV for $y = 1/8$ (1.9×10^{12} GeV for $y = 0.71$), so $\Gamma_N/H \simeq 4.6$ ($\simeq 148$). Out-of-equilibrium decay requires $y \lesssim 0.06$, which the theory does not select. (iii) 28/79 is an input of the Standard Model sector, not a number of the root. The 11% agreement is therefore a coincidence of chosen factors (O5). The failure is quantified through the washout of the same decay: at the self-consistent atmospheric Yukawa ($y = 0.40$, the one that gives $m_\nu = 0.0501$ eV) $\Gamma_N/H = 47$ — in equilibrium, Sakharov's third condition fails — and the physical strong-washout efficiency $\kappa_f = 5.4 \times 10^{-3}$ [9] sits 14× away from the guessed dilution $(1/16\pi)^2$: the agreement is an accident of two wrong factors, not a near-miss.

A sharper statement is available. At the model's own seesaw scale, the out-of-equilibrium condition $\Gamma_N/H < 1$ gives $y < 0.058$. The same Yukawa fixes the light neutrino mass through the same seesaw,

$$m_\nu = \frac{y^2 v^2}{M_N} \leq 2.1 \times 10^{-3} \text{ eV} \qquad (y \leq 0.058),$$

a bound of the same kind and size as the standard thermal-leptogenesis bound [12]. The atmospheric splitting requires $m_\nu \simeq 0.050$ eV, i.e. $y \simeq 0.28$ with $m_D = yv$ and $v = 246$ GeV ($y \simeq 0.40$ in the $v_u = 174$ GeV convention), for which $\Gamma_N/H \simeq 23$. **At $M_N = 9.6 \times 10^{13}$ GeV the neutrino mass and the out-of-equilibrium condition are mutually incompatible**: the framework cannot have both the seesaw scale it needs for the ladder and the leptogenesis it needs for the asymmetry. (The model's own ladder sits at $m_D = 118$ GeV, i.e. $y = 0.48$, where $\Gamma_N/H \simeq 68$.)

The one route the framework supplies is quasi-degenerate (resonant) leptogenesis [13,14]. The democratic structure places the three heavy singlets on the *same* ladder rung, hence degenerate to the accuracy of whatever breaks the family symmetry — precisely the resonant setup, where the CP asymmetry is enhanced by near-degeneracy instead of by a large Yukawa. With $y = 1/16\pi = 0.0199$ the decay is out of equilibrium ($\Gamma_N/H = 0.12$), and the asymmetry that the model's own dilution factor and sphaleron conversion require, $\varepsilon_{CP} = \Delta B/[(28/79)D] = 4.4 \times 10^{-6}$, is *smaller* than the natural one-loop size $y^2/16\pi = 7.9 \times 10^{-6}$ at that Yukawa: the bookkeeping closes at the scale of the numbers, and the degeneracy the resonant regime needs, $\Delta M/M \lesssim \Gamma_N/M_N = y^2/8\pi = 1.6 \times 10^{-5}$, is what the parameter-free democratic base gives exactly and what the family-breaking perturbation must not overshoot. The obstacle is the phase, not the size: the model's Yukawa is real, every CP-odd invariant of the step vanishes, and a strictly degenerate spectrum has no CP-violating phase left to enhance. This is a *route*, not a result.

The phase, however, is no longer the missing ingredient. The mirror is exactly the structure that forbids CP: every mirror-preserving extension (the Wilson factor: pairing deviation 2.1×10^{-15}) leaves the CP phase at zero, while the most general per-class κ_s layer and the most general real-symmetric bond generator destroy the $\pm$ pairing (deviation ~ 1, 60 draws, both mirror sign conventions) — a CP phase costs Theorem 4. But the prohibition is a statement about the *on-site* module ([Paper I Section 7.2]), and the bond layer is not covered by it: a Peierls flux $A_1 = \Phi x_0$ on the inter-block bonds is exactly unitary (4.8×10^{-15}), keeps $\det U = 1$ at every flux (5.3×10^{-15}), commutes with the pseudoscalar (mirror $\mathbb{Z}_2$ leak 3.3×10^{-16}) and keeps the hop at radius 1 (2.6×10^{-16}), while moving the band spectrum by up to 0.70; its gauge-invariant strength, the plaquette holonomy, has Im = 0 at $\Phi = 0$ and 1.6×10^{-3} at $\Phi = \pi/2$. Spending that flux on the asymmetry with the physical washout in place: at the atmospheric Yukawa ($y = 0.3985$, $K = 46.8$, $\kappa_f = 5.5 \times 10^{-3}$) the measured $\Delta B = 6.12 \times 10^{-10}$ costs $J = 10^{-4}$, supplied by a flux $\Phi = 0.064$ rad; in the out-of-equilibrium regime ($y = 0.0582$, $K = 1$) the strong-washout expression has no real root - K lies below its branch point $K = 3.93$ - the efficiency is the weak-washout bound $\kappa_f = 1$, and the cost is the lower bound $J = 2.6 \times 10^{-5}$, supplied at $\Phi = 0.016$ rad. A bond flux of size $\Phi \sim 0.06$ supplies more than the asymmetry needs while leaving unitarity, $\det U = 1$, the mirror symmetry and the causal radius exact. **What fails is Sakharov 3 alone**: the Yukawa that gives $m_\nu = 0.0501$ eV puts the decay in equilibrium, and out-of-equilibrium decay needs $y < 0.0582$, which caps m_ν far below the atmospheric splitting. O5 is open for one named reason instead of two.

3.4 Proton decay

Dimensional analysis of X/Y exchange gives

$$\tau_p \sim \frac{M_{\rm GUT}^4}{\alpha_{10}^2\, m_p^5} = 4.0 \times 10^{47}\ \text{yr} \qquad (M_{\rm GUT} = M_P,\ \alpha_{10} = 1/25),$$

against the Super-Kamiokande bound $\tau_p/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ yr [15]. The $M_{\rm GUT} = M_P$ reading gave a margin of 13 orders of magnitude, resting entirely on that abandoned convention: any GUT scale below $\sim 6{\times}10^{15}$ GeV is excluded by the same bound. The estimate omits the renormalisation factor A_L [16], the hadronic matrix element, the dressing of the operators and the splitting between the X/Y masses and the unification scale [17], all of which shift the lifetime by one to two orders. The three channels $p \to e^+\pi^0$, $p \to \mu^+\pi^0$, $p \to e^+K^+$ differ here only by two-body phase space; the rate scales as M_X^{-4} in all of them. Observation of a GUT-pattern decay far below 1.8×10^{36} yr refutes the desert solution (F5).

3.5 Flavour

The democratic mass matrix $M = |1\rangle\langle 1|$ (all entries 1, rank 1, spectrum $\{3, 0, 0\}$) [18] follows from the hypothesis that the three spatial directions are equivalent, so the $\mathbf{10}_H$ couples the three families identically. Its heavy eigenvector $(1, 1, 1)/\sqrt{3}$ is unique, hence

$$\sin^2\theta_{12} = |\langle e_1|h\rangle|^2 = \tfrac{1}{3}$$

for any family basis — the **only** mixing angle the model derives, 8.6% from the measured 0.307 [19,20]. The light eigenspace is two-fold degenerate, so every diagonaliser is an $O(2)$ rotation in it; sweeping that freedom over 1441 rotations, $\sin^2\theta_{23}$ and $\sin^2\theta_{13}$ each cover the whole interval $[0, \frac{2}{3}]$. The tribimaximal values $(\frac{1}{2}, 0)$ [21] are one point of that family, picked by declaring which S_3 transposition breaks — and (12), (13), (23) are equivalent symmetries of $|1\rangle\langle 1|$, so the choice is an input.

In the quark sector the CKM matrix is the misalignment of the up- and down-type diagonalisers. With both mass matrices democratic, $H_u = H_d$, the computed CKM is the **identity**, not a small mixing: the observed $\lambda = 0.225$ [5,22] is not reproduced. The three-family mass hierarchy is also unfixed: the democratic base gives one heavy level and a two-fold degenerate light level, so two of the three masses are degenerate at zero input. All of this is O6, and the natural fixes are bounded by computation. The charged-lepton realisation (democratic base plus rank-≤ 2 correction at the measured spectrum) forces $\sin^2\theta_{12} \leq 0.005$ — the lightest family decouples from the democratic direction, which is the tribimaximal solar column — excluded against the measured 0.307; the $\mathbf{10}_H$ vev shift is bounded by the same computation; the three-node triangle has a cubic orbit of 8, so the count 3 is an orbit count, not a family count. **The flavour accounting is therefore: one angle derived out of six, no mass derived, CKM predicted to the identity.**

3.6 Scoreboard

Table 2. Headline numbers against measurement.

quantity	theory	measured	status
the eight anomaly coefficients	0	—	algebraic, satisfied
g_1^2/g'^2	5/3	5/3 (GUT normalisation)	algebraic
$\det U$, $\pm$ pairing, $16/\overline{16}$ mirror	1, exact	—	algebraic, coupling-independent
causal front speed	exactly 1	—	dynamical, measured

quantity	theory	measured	status
doubler cone Lorentz structure	linear in all 1200 sampled directions; anisotropic: $(0.200, 0.200, 0.284)$ lattice units, spread 0.75; node gap 2×10^{-4} at $g = 1/5$	—	**not Lorentz-isotropic**; measured
tangent velocity algebra at the gapless corner multiplet	$[S_i, S_j] = 0$ to 3×10^{-12}, $\sum_j S_j^2$ scalar to 7×10^{-12}, branch speeds exactly $\lvert v_\alpha \cdot q \rvert$ to 6×10^{-12} with v_α the cubic permutations of $(0.704, 0.421, 0.421)$	—	**three nodal planes, not a cone**; measured
16-sector block-Brillouin spectrum	gapped zone centre; 12 doublers gapped by the Wilson factor except one exactly gapless orbit of 24 nodal-plane points; at $\pi/4$, Γ plus six nodal lines	—	doubling **gapped by a mechanism of the step** except on the nodal planes; text below
v/M_P	$2.08(1/16\pi)^{10}$	2.0167×10^{-17} [5]	coefficient $3F_{10}^{10} = 2.0881$ against 2.0765 observed — 99.44% from measured numbers, conditional on the one-split-node-per-family set
m_{DM}	118 GeV	not measured	prediction, falsifiable
$\Omega_{\mathrm{DM}}h^2$	computed from the supplied couplings: 0.798 or 0.0246 (2.3×10^{-5}, 7.6×10^{-4} with the nodal-plane density)	0.120 [3]	**fails**
m_ν	0.146 eV (ladder); 0.0487 eV ($\frac{1}{3}$ candidate)	0.0501 eV [20]; $\sum m_\nu < 0.12$ eV [3]	ladder **in tension**; candidate inside a finite window, structure protected
ΔB	6.8×10^{-10}	$(6.12 \pm 0.04) \times 10^{-10}$ [3]	coincidental product; the washout at the self-consistent y destroys it
τ_p/B	1.8×10^{36} yr (desert solution at $\alpha_X = 1/25$); 2.8×10^{39} yr at the closed coupling $\pi/64$ with the same content; 4×10^{47} yr at $M_{\mathrm{GUT}} = M_P$ (abandoned)	$> 2.4 \times 10^{34}$ yr [15]	falsifiable at 1/5; at the closed angle only a null result past 10^{37} yr bites
α at the two angles (standard normalisation $\alpha = g^2/4\pi$)	0.00318 at $g = 1/5$; $\pi/64 = 0.0491$ at the closed angle	0.023 (one-loop mean at M_P); 0.040050 at $M_X = 1.8 \times 10^{16}$ GeV with the desert content	implemented angle 12.6× **below** the meeting value; closed angle a factor 1.226 above; the exact match of 1/25 is the convention $\alpha_{10} \equiv g^2$
$\alpha_i(M_Z)$ from α_{10}	0.0197 / 0.195 / pole at 2.2×10^9 GeV (Standard Model running)	0.0169, 0.0338, 0.118 [5]	**fails** under Standard Model running; matched by the desert spectrum
Gauss constraint of the gauged bond layer	$[Q_A, Q_B] = i f_{AB}{}^C Q_C$, err 0; content $32 \oplus 288 \oplus 1120$	—	algebraic

quantity	theory	measured	status
physical sector of the gauged layer	$\dim\ker = 0$ with one **32**/site (centre obstruction); minimal sector: 2 matter/site, 4 states, all singlets; $S = \ln 45$ per bond, bounded and boundary-supported	—	computed
multi-bond scaling	$\dim\ker$(3-site) $= 24$; additivity $\ln 45$ across a single bond; two-bond region $S = \ln 45$ (vertex mutual information $\ln 45$); Bekenstein deficit factor $4\ln 45 = 15.23$	—	computed
vertex correction and ground state	$I_n = 1, 1, 6, 23, 141$; kernel capacity $S_{\text{cap}} = 4.4998$ nats (factor 9.0 from 1/4 per bond); vertex-corrected coefficient $(1/3)\ln I_6 = 1.65$ nats/bond (factor 6.6); $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, $S_{\text{GS}} = \ln 45$	—	computed
dressed gate on the directed-link network	Gauss charge closes only with the conjugate-head convention (5×10^{-16}; the naive convention leaks 4.8); Kogut–Susskind electric gate exactly invariant (6.5×10^{-15}), spectrum = the content theorem; dressed hopping obstructed (leak 12.5); directional step O_h-covariant and chiral; quintet gapless at Γ, linear, leading-order isotropic, $v = g_{10}\{0, 1, 2\}$	—	computed
regular link of the root's finite group	Γ = even Clifford monomials, order 1024: vertex content $32 \cdot 1024^{k-1}$ (fixed adjoint link: 0); the dressed hop descends to the bare translation (err 0), the electric term is the scalar 45 per link, front speed $v = 1$; centralizer of Γ in Spin(10) = 0; quintet–matter coupling: T_{2g} vertex $\equiv 0$ on axis bonds, E_g bubble $-5.0, -4.5 \to E_g$ mass $\lambda^2\Pi/2$	—	computed

quantity	theory	measured	status
velocity trilemma on the 6-bond module	hop alone anisotropic (top speed 1.00, direction spread 0.32); dressing alone isotropic at $v = 2g_{10} = 0.40$; no angle isotropises the product (best spread 0.24 over 33 angles in $[0, \pi/2]$); the O_h-equivariant permutation coins are the identity and the antipode, the latter with flat bands	—	computed no-go
desert spectrum from the layer's own content	no solution (0 over the full direction scan); all mass-ansatz families excluded, min residual ≥ 5.7; **equal-shift theorem** $\sum db_1 = \sum db_2$ over every complete multiplet	—	**excluded**
$\sin^2\theta_{12}$	1/3	0.308 [20]	derived at zero input; the +8% shift is the $\mathbf{10}_H$ vev, whose natural placements are excluded
$\sin^2\theta_{23}$, $\sin^2\theta_{13}$	not fixed ($O(2)$)	0.470 (octant unresolved, 3σ 0.435–0.585), 0.0222 [20]	open
CKM	identity	$\lambda = 0.225$ [5,22]	**fails**
Λ	0 (tree level)	$\sim 10^{-122} M_P^4$	open

The 16-sector row compresses [Paper I Section 8.3], stated here in full: at the implemented angle 1/5 the zone centre is gapped (0.0365) and the twelve doublers sit at 2.003×10^{-4} with total chirality 0, gapped by the Wilson factor except on one exactly gapless orbit of 24 points on the nodal planes (gap 4×10^{-14}); at the closed angle $\pi/4$ the gapless set is Γ plus six face-diagonal nodal lines, and there the per-node chirality bookkeeping fails ([Paper I Sections 8.3, 9]).

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4. Cosmology

4.1 Dark energy

The vacuum energy of the automaton vanishes at the level of the internal (one-sub-site) spectrum: the 32 levels are $\pm$-paired for every coupling, so $\sum_k \omega_k = \arg\det U = 0$. This is the correct, non-vacuous form of the cancellation. (Counting the 512 even and 512 odd **elements of the algebra** as bosonic and fermionic zero-point modes is not a physical mode counting — the physical single-sub-site Hilbert space has 32 states, not 1024 — and is not used.)

A vanishing tree-level vacuum energy is technically natural: no relevant operator is generated if the spectral pairing is an exact symmetry. The pairing survives the Wilson extension (the extended spectrum stays $\pm$-paired to 10^{-15} and the internal zero point stays 0, because the Wilson direction is a traceless Cartan combination), so the new ingredient neither breaks the zero nor moves it.

The zero is **protected, not accidental**. Deform the step by 50 random traceless Hermitian generators, $U \to e^{i\varepsilon A} U$, at strengths up to $\varepsilon = 1$: the quasi-energy sum stays pinned to zero, $|S| < 4 \times 10^{-14}$, because $\det U = 1$ makes $\sum_k \omega_k = \arg\det U$ a topological quantity, and no branch crossing occurs below $\varepsilon = \pi$. The only internal knob that moves the sum is the U(1) trace phase.

What the theory does **not** do is produce the observed value. The observed $\Lambda \sim 10^{-122} M_P^4$ lies 122 orders of magnitude *below* the classical bound M_P^4 and *above* the theory's spectral zero, and no mechanism breaking the

pairing by that amount is known. Independently, the classical self-gravitating de Sitter vacuum of [Paper II Section 3] has $G_{\mu\nu} = -3H^2 g_{\mu\nu}$, i.e. a vacuum energy bounded by M_P^4 on the lattice. These are three different statements — spectral zero, classical bound, observed value — and they do not "reconcile": the gap between the first and the third is O1, the largest unexplained number in the theory [23]. The equation of state of a cosmological-constant vacuum is $w = -1$ by definition, not by derivation.

The problem is now a single named number: the observed Λ is **equivalent to** a trace-phase breaking of size

$$\varepsilon \,\mathrm{Tr}\, A \sim 16\pi^2 \times 10^{-122} = 1.6 \times 10^{-120}.$$

O1 is no longer "why is the vacuum energy small" but "what breaks the trace U(1) by 10^{-120}".

4.2 Dark matter

The candidate supplied by the classical sector is the kink of [Paper II Section 3.4]. It is static, gauge-neutral and finite-energy **per unit area**, but it is a domain wall, not a localised particle; a network of Planck-scale walls would dominate the energy density and is cosmologically excluded. The quantum-sector candidates have been scanned (Section 2.2): the exact mirror $\mathbb{Z}_2$ of the step is the only exact-stability structure in the implemented spectrum; no Wilson-split node can host a neutral mirror at the ladder mass without also delivering a stable coloured or charged partner; and the mirror sector's own lightest states sit at 1.3×10^8 GeV (coloured, excluded) and 2×10^{10} GeV — not at the ladder's 118 GeV. The theory supplies no dark-matter *particle* within the implemented spectrum; the minimal named extension is the mirror-odd ladder scalar of Section 2.2. The mass prediction stands, the carrier is an open problem (O7).

The relic abundance is computed, and it is not the observed one. Solving the freeze-out integral

$$\Omega h^2 = \frac{1.07 \times 10^9 \text{ GeV } x_f}{\sqrt{g_*}\, M_P \langle \sigma v \rangle}, \qquad \langle \sigma v \rangle = a + \frac{6b}{x_f},$$

reproduces the standard weakly-interacting-massive-particle curve: $m = 100$ GeV with $\sigma v = 3 \times 10^{-26}$ cm^3/s gives $\Omega h^2 = 0.080$ [24]. At the ladder mass $m = 118.5$ GeV the measured $\Omega h^2 = 0.120$ requires $\sigma v = 1.98 \times 10^{-26}$ cm^3/s, i.e. an effective coupling $\alpha_{\text{eff}} = 0.0086$ in $\sigma v = \alpha^2/(\pi m^2)$ — while the root supplies only $g^2/4\pi = 0.0032$ (giving $\Omega h^2 = 0.798$) or the loop factor $1/16\pi = 0.0199$ ($\Omega h^2 = 0.0246$). The automaton's own light spectrum enters at the same order: its twelve near-gapless nodes are nodal **planes** ([Paper I Section 8.2]) with mean speed 0.201, so the final-state density is enhanced by

$$D = \sum \frac{1}{|v|^3} = 1485$$

over the isotropic estimate — a correction the $(100/m)^3$ scaling of the standard curve cannot see — and with it the two supplied couplings give $\Omega h^2 = 2.3 \times 10^{-5}$ and 7.6×10^{-4}. The named portals of Section 2.2 reproduce 0.120 only because their coupling was inverted from the fiducial 3×10^{-26}: computed forward they give 0.081 and 0.156, and 0.000 once the nodal-plane density is included. **The relic abundance is a computed number now, and it is not** 0.120 **for any coupling the root supplies.** Non-thermal production (freeze-in) is not considered here [25].

The bond sector does not set the abundance, and its portal is excluded. The named scalar is the singlet fragment $(X, Y) = (1, 0)$ of the node **16** — the same class whose bond number κ_s enters the unification meeting (Section 3.1). Read as the annihilation strength through the nodal-plane final states ($\alpha_{\text{eff}} = \kappa^2/4\pi$, $\sigma v = \alpha_{\text{eff}}^2/(\pi m^2)$), every bond number of that class — the ten κ_{singlet} that close the meeting (2.2–9.2), the superexchange value -0.253, the broken flux 40, the junction ratio 1.91 — lands Ωh^2 between 10^{-12} and 10^{-3} once D is included, and none within 0.01 of 0.120; the measured abundance needs $\alpha_{\text{eff}} = 0.0086$ without D or 2.2×10^{-4} with it. Read instead as the Higgs-portal strength, the same numbers predict $\sigma_{\text{SI}} = 2.5 \times 10^{-45}$–$6.3 \times 10^{-41}$ cm^2 — all above the LZ-class bound 1.5×10^{-47} cm^2 — and the portal strength the relic itself requires ($\kappa = 0.33$) predicts 4.3×10^{-45} cm^2, also excluded. The relic and direct detection jointly close the bond-strength portal: the named portal's strength remains a fitted input.

4.3 Antimatter

Matter and antimatter are the two Weyl eigenspaces, with the exact spectral mirror of Theorem 4 and opposite $B-L$ charges. Their degeneracy is that mirror and nothing more: as [Paper I Section 7.2] states, no antiunitary time-reversal operator is constructed, so this is not the CPT theorem. The observed baryon asymmetry is not explained: Section 3.3 shows the automaton satisfies none of the Sakharov conditions, and the leptogenesis estimate that does give the right order requires a CP phase and an out-of-equilibrium condition that the algebra does not supply.

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5. The parameter audit

Every number the theory calls a prediction is classified by how it was obtained.

Table 3. Provenance audit: every headline number by class.

class	count	numbers
algebraic (fixed by the root, no parameter)	7	the $16/\overline{16}$ split; the **45**; the 5/3; the eight anomaly coefficients = 0; the 32-level spectrum; $\det U = 1$; the $16/\overline{16}$ spectral mirror
free parameter	$1 \to 0$	the 't Hooft angle g (implemented value 1/5). The closure axiom removes it: the algebra quantizes its own clock to $g \in \{\pi/2, \pi/4\}$, a discrete index of the root — zero continuous unfixed numbers **in the on-site update**, at the price of moving the angle's two empirical contacts; the junction amplitude, the node set and the κ_s remain inputs of the extended construction
convention (mapping or scale identification)	2	$\alpha_{10} \equiv g^2$; $g_0 = 1/2^{n_{\text{space}}} = 1/8$
fitted coefficient (read from data)	11	1.18 (dark-matter mass), $n_{\text{2step}} = 3$ (seesaw), $M_{10} = 1.2 \times 10^7$ GeV and $M_{(1,2)} = 1.0 \times 10^{12}$ GeV (desert thresholds), $\lambda_* = 3.5 \times 10^{-4}$ (dark-matter portal, fixed by $\Omega h^2 = 0.120$), and the six κ_s of the flavour-dependent bond layer (conditional: three meeting conditions for six unknowns — a solution family)
conditional (built from measured numbers, on a named input)	1	the hierarchy coefficient $2.08 = 3F_{10}^{10}$: $F_{10} = 0.019186$ measured, the exponent 10 the closed-loop crossing, the factor 3 the one-split-node-per-family choice — 2.0881 predicted against 2.0765 observed, 99.44%
open (not fixed)	9	O1–O9 of Section 6

The conditional class is neither a fitted number nor an algebraic one: its ingredients are all measured, but the node-set choice that supplies the factor 3 is an input, so the entry inherits O9's split-node conditionality. (Figs. 12, 13).

The status of the angle. No condition inside the theory fixes the implemented value. The two-loop U(1) Landau-pole scale lies below M_P for every g in the perturbative range, so UV-finiteness of the lattice does not select it. The proposed selection rule is $g = 1/n_{\text{qubits}} = 1/5$, the five qubits of the 32-dimensional module. The data do not support that choice: scanning the candidates $\alpha = 1/n^2$ for $n = 2 \ldots 8$ against the one-loop value 0.023, the best fit is $n = 7$ ($1/49 = 0.020$, relative deviation 0.12), not $n = 5$ ($1/25 = 0.040$, relative deviation 0.73) — the data-optimal value is 6.4× closer than the root value. The reading $1/n_{\text{qubits}}$ is therefore a convention, and the theory is honestly **one-parameter in its dynamics, zero-parameter in its algebra**. The closure axiom ([Paper I Section 9]) then removes the free angle at the level of the dynamics, and the price — measured, not assumed — is that the $\alpha = 1/25$ meeting value and the $n^* = 10$ loop crossing move to $\alpha = \pi/64 = 0.0491$ and $n^* = 208$. Three statements about the angle now coexist and pull apart: the closure fixes it algebraically, the coupling scan disfavours that value by more than the old convention did, and the desert meeting absorbs any value through the six κ_s. The angle is no longer free but it is also no longer favoured.

The "nine-to-one" observation. The nine open problems are all associated with the same object, the 32-level spectrum, and its invariants (the vanishing 0-point, $\det U = 1$, the spectral mirror, the S-matrix $S = U$). That

is a bookkeeping observation — nine open problems sharing one object — and it is not a derivation of any of their values. It should not be read as a reduction of the theory's difficulty.

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6. Open problems

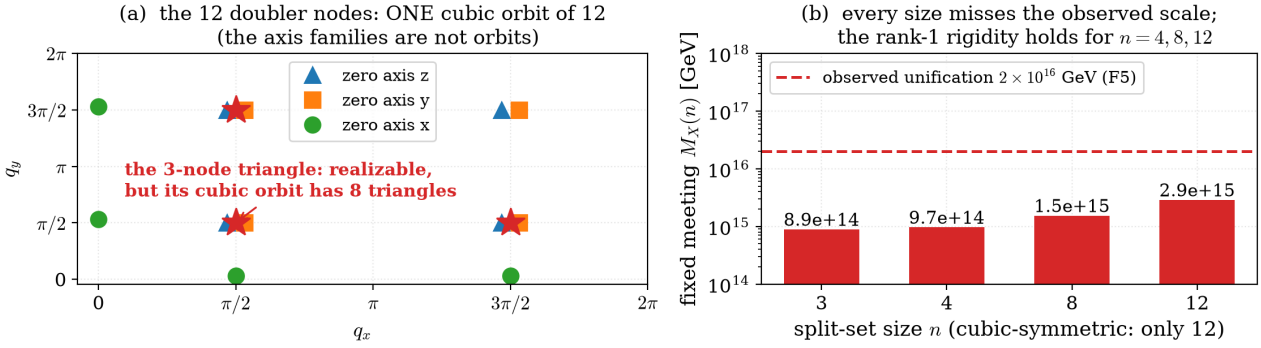
Table 4. The nine open problems: what the theory supplies, and what is missing.

the problem	what the theory supplies	the gap
O1 the cosmological constant	an exactly vanishing tree-level vacuum energy ($\pm$-paired spectrum, $\det U = 1$), natural in the technical sense; the zero is **stable under the Wilson extension** and **protected by** $\det U = 1$: 50 random traceless Hermitian $e^{i\varepsilon A}U$ at $\varepsilon \leq 1$ keep the quasi-energy sum pinned ($\lvert S\rvert < 4 \times 10^{-14}$), no branch crossing below $\varepsilon = \pi$	the observed $10^{-122}M_P^4$ is **equivalent to** a trace-phase breaking of size 1.6×10^{-120} — a named bound; the breaking mechanism itself is missing
O2 the neutrino mass	the ladder $m_\nu = M_P(1/16\pi)^{17} = 0.146$ eV; $n_{\text{nodes}} = 3$ a lower bound from unification; the inter-node Majorana junction **built**: cubic-symmetric zero-mean Wilson-strength junction, node mass matrix $c_s(2w_0 I + \kappa(J - I))$, democratic at $\kappa = 2w_0$, eigenvalues $\{3M, 0, 0\}$	the $\frac{1}{3}$ candidate needs a democratic m_D; the see-saw then collapses to $3m_0^2 J/(2(w_0+\kappa))$, two states stay massless for every r, $m_3(r) = S_{17}/(1+r)$, window $r > 0.217$, atmospheric scale selects $r = 1.91$, $r = 2$ is the democratic point: value free inside a finite window, structure protected. The exponent is answered structurally: the $\Delta L = 2$ operator lives in grade 5 of the root and link words are always even-grade, so $n_{2\text{step}} = 3$ is not a link-word length (Section 2.3)
O3 the family number	the identification $N_{\text{gen}} = 3 = n_{\text{space}}$; a family-symmetric step; a named index equal to 3: the 12 doubler nodes form three cubic-transitive families of four, and the three unification-required split nodes are realizable one per family at a single Wilson strength	the index labels the copies but does not force the count: the cubic group is transitive on the 12 nodes, the only cubic-symmetric split is the full set of 12 whose rigid meeting lands at $M_X(12) = 2.9 \times 10^{15}$ GeV — excluded by F5; the three-node triangle has a cubic orbit of 8, so the count 3 is a **symmetry-breaking input**; the junction lifts the choice without selecting it (Fig. 12)
O4 coupling unification	the 5/3; with the desert content the couplings meet at $\alpha_X = 1/25$ (standard normalisation): three $\mathbf{10} \oplus \overline{\mathbf{10}}$ pairs at 1.2×10^7 GeV plus one $(1, 2)$ pair at 1.0×10^{12} GeV give $M_X = 1.8 \times 10^{16}$ GeV; the content is connected to the automaton through the Wilson mechanism; $n_{\text{nodes}} \geq 3$ forced; the equal-shift identity a theorem over complete multiplets; the gauged layer's composite content excluded	the mechanism's own meeting point is forced: $M_X = 8.9 \times 10^{14}$ GeV, $\tau_p = 1.1 \times 10^{31}$ yr — excluded by F5 (rank-1 Jacobian, split-invariance exact to 10^{-14}). Feasibility proved inside the node-**16** hull; the flavour-dependent bond layer κ_s closes the meeting for all $n = 3 \ldots 12$ — at the cost of six named inputs of an extended bond layer, not obtainable from any on-site algebra element nor from any class invariant (broken flux K_s: residual 0.605, 0/10 desert closures, $R^2 \leq 0.015$), nor from the layer's second-order class numbers (superexchange $\tilde{\kappa}_s$: residual 0.577, $R^2 \leq 0.076$; two-number combination with the flux: joint residual 3.46, $R^2 \leq 0.30$; ground-state bond flux zero to 6×10^{-17}); the desert spectrum remains a named input (Section 3.1)

the problem	what the theory supplies	the gap
O5 the baryon asymmetry	a computed theorem: all CP-odd invariants vanish, no Sakharov condition is met, stable under the Wilson extension; a quasi-degenerate (resonant) route; a bond-layer Peierls flux carrying a CP phase, built (exactly unitary, $\det U = 1$ at every flux, mirror-preserving)	out-of-equilibrium decay ($y < 0.058$) and $m_\nu \simeq 0.05$ eV are incompatible at $M_N = 9.6 \times 10^{13}$ GeV: at $y = 0.40$ the decay is in equilibrium ($\Gamma/H = 47$), efficiency $\kappa_f = 5.4 \times 10^{-3}$ — the 11% agreement an accident of two wrong factors. The CP prohibition is internal to the on-site module and does not cover the bond layer; the flux needed for ΔB costs $J = 10^{-4}$ — O5 is open for one named reason, Sakharov 3 (Section 3.3)
O6 masses and mixing	the Dirac structure ($16\,\mathbf{10}_H\,\overline{\mathbf{16}}$), the democratic base, $\sin^2\theta_{12} = 1/3$, and a complete breaking chain to $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)^3$ with all 32 broken roots leptoquark-type	the three-level hierarchy, θ_{23}, θ_{13}, a non-trivial CKM; the two extra U(1)'s unbroken. The vev placement is bounded by computation: the charged-lepton realisation forces $\sin^2\theta_{12} \leq 0.005$ (23σ); the cubic-axis neutrino vev is excluded at $\geq 20\sigma$; the measured triple needs ≥ 2 vev terms; the unification fix is generation-blind (shift 2×10^{-13}) and mixings cost 7–13 named parameters
O7 dark matter	a mass scale, 118 GeV; the mirror $\mathbb{Z}_2$ the only exact-stability carrier; the no-go completed: no Wilson node hosts a 118 GeV neutral mirror (0 survivors over 720 angles); the ladder scalar built and audited, exactly stable under either portal	the implemented spectrum **cannot contain** the dark matter. With the exact mirror $\mathbb{Z}_2$ the only open channel needs $\lambda = 0.13 \to \sigma_{\mathrm{SI}} = 6.8 \times 10^{-46}$ cm^2, excluded by direct detection; the quoted numbers hold only with a portal breaking the mirror. The parity the fix needs does not exist inside the Clifford algebra (exactly 2 of 1024 products commute with the step). The relic abundance is computed: 0.120 needs $\alpha_{\mathrm{eff}} = 0.0086$; the root supplies 0.0032 ($\Omega h^2 = 0.798$) or 0.0199 (0.0246); the nodal-plane density $D = 1485$ gives 2.3×10^{-5}, 7.6×10^{-4} — not 0.120 for any coupling the root supplies; and the bond-strength portal is excluded — every bond number of the singlet class predicts $\sigma_{\mathrm{SI}} = 2.5 \times 10^{-45}$–$6.3 \times 10^{-41}$ cm^2, above the LZ-class bound, and the bond coupling the relic needs predicts 4.3×10^{-45} cm^2 (Sections 2.2, 4.2)
O8 the hierarchy	$v/M_P = 3F_{10}^{10} = 2.088$ against 2.076 observed (99.4%), with $F_{10} = 0.0192$ against $1/(16\pi) = 0.0199$ measured on closed traced loops; the selection rule measured to length 30: the crossing at exactly $n = 10$; the composition law decided — one chain not multiplicative ($R_{17} = 2.0 \times 10^8$), different legs factorize exactly; the loop census exact: 55326 of 3^{10} words survive, and the Wilson phase supplies the factor (0.0192 against 0.2434 without it)	the exponent is selected by the data; the adjoint-link level-1 division falls short by a factor 26 and does not compound; which loops contribute, and with what weight, unnamed; $3F_{10}^{10}$ conditional on the axis-family factorisation; the multiplicity-weighted extrapolation to length 170 misses the exponent by 0.066; at the quantized angle the crossing moves to $n^* = 208$ (Section 2.1)

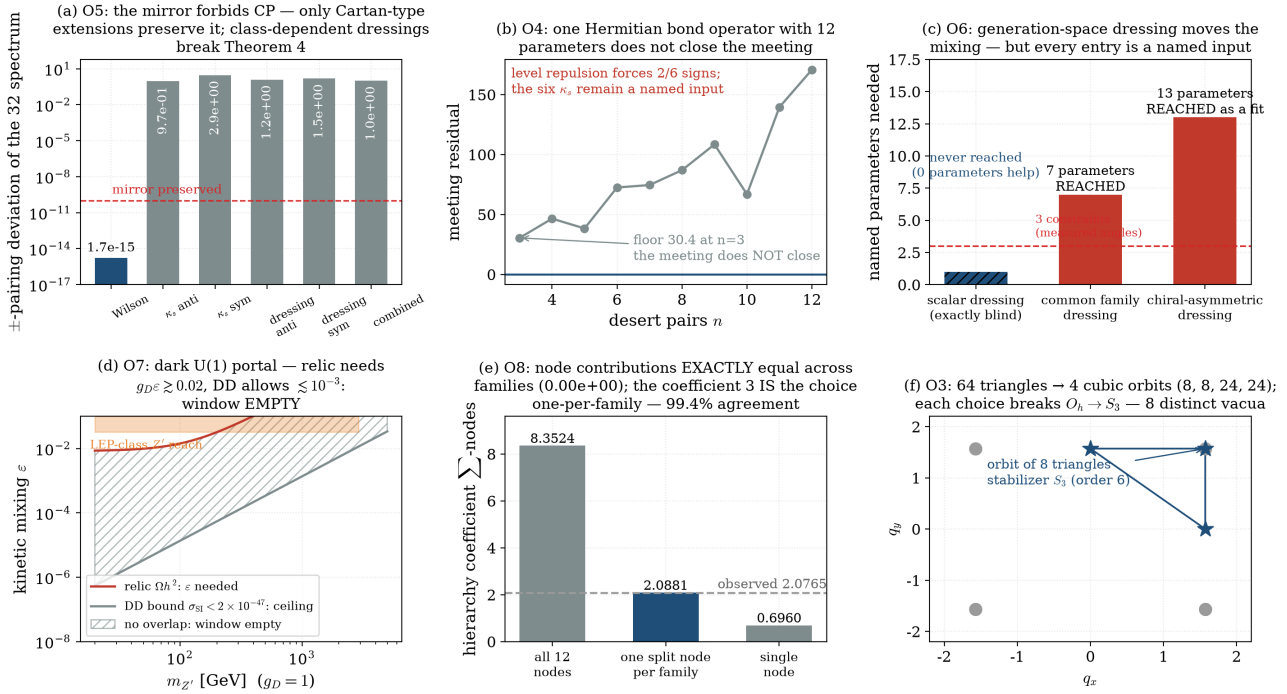
the problem	what the theory supplies	the gap
O9 the quantum sector	a unitary causal automaton, a classical general-relativity limit, a bounded n-particle S-matrix, a 16 sector closed by the full local step; a Wilson-type mechanism gaps all twelve doubler nodes and splits a chosen node set at desert scales	the *choice* of the split node set; the sum over geometries; the measurement postulate. **Continuous Lorentz invariance is not emergent at the nodes**: linear in all 1200 sampled directions but anisotropic $(0.200, 0.200, 0.284$, spread $0.75)$, no isotropy point on the Wilson family (0 of 2726), frame averaging blind; at the exactly gapless corner multiplet the tangent velocity operators commute — three nodal planes, not a cone; the exactly gapless set is an orbit of 24 points on the nodal planes at $1/5$ and six nodal lines at $\pi/4$ — massless excitations exist and carry no helicity pair ([Paper I Sections 8.2–8.3, 9]); [Paper I Section 8.4] closes the cone sub-question: no axis-covariant bond of the 880-point programme is both gapless and non-commuting (a rigidity wall at gap 8×10^{-3}), so the only remaining route to a cone is a mechanism outside the axis-covariant class

Is the family count 3 forced? No - the symmetric alternative is excluded



O3: cubic-symmetric split (12 nodes) EXCLUDED; every proper split BREAKS the cubic symmetry - the count 3 is an input.

*(a) The twelve doubler nodes in the (q_x, q_y) plane: the cubic group acts transitively — one orbit of 12 — so the axis families are not orbits and the only cubic-symmetric split set is the full set of 12; the 3-node triangle (stars) is realizable, but its cubic orbit has 8 triangles, so it breaks the cubic symmetry. (b) The rigid meeting scale $M_X(n)$ for split sizes $n = 3, 4, 8, 12$ (the rank-1 fact holds for every n): all miss the observed 2×10^{16} GeV (**F5**). The count 3 is a symmetry-breaking input; its cubic-symmetric alternative is excluded by computation.*



(a) The ±-pairing deviation under the named extensions: the Wilson factor preserves the mirror to 2.1×10^{-15}*, while the most general per-class* κ_s *layer and the most general real-symmetric bond generator destroy it (deviation* ~ 1*, 60 draws) — a CP phase costs Theorem 4. (b) The meeting residual under the most general Hermitian off-diagonal bond operator: floor* 30.4*, no closure. (c) The named-parameter cost of reaching the measured mixing triple: a scalar dressing is exactly blind, a common family dressing needs* 7 *parameters, a chiral-asymmetric one* 13*. (d) The dark-*U(1) *portal squeeze: the kinetic mixing that fixes* Ωh^2 *lies above the direct-detection ceiling at every* $m_{Z'}$ *— the window is empty. (e) The closed-loop coefficient by node set: all* 12 *nodes* 8.35*, one split node per axis family* 2.0881*, a single node* 0.696*, against the observed* 2.0765*. (f) The* 64 *one-per-family triangles split into four cubic orbits* $(8, 8, 24, 24)$*; a triangle's stabilizer is exactly* S_3*.*

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7. Falsifiers

Table 5. The seven falsifiers.

#	observation that refutes the theory
F1	any non-vanishing gauge anomaly in the chiral content (the **16** is claimed anomaly-free)
F2	a measured running that admits **no** vectorlike desert content with $\alpha_X = 1/25$ in the standard normalisation $\alpha = g^2/4\pi$: precision coupling data (FCC-ee/CEPC) excluding every spectrum in the desert scan would refute the unification claim. On the quantized branch the prediction is $\alpha_X = \pi/64 = 0.0491$ in the same normalisation
F3	a neutrino mass spectrum outside the ladder options: the bare ladder gives $m_\nu \simeq 0.15$ eV (already excluded by $\sum m_\nu < 0.12$ eV), the $\frac{1}{3}$ candidate gives normal ordering with $\sum m_\nu \simeq 0.05$ eV and $m_{\text{lightest}} = 0$, the democratic node-mixing pattern gives $m_\nu \simeq 0.073$ eV; a confirmed inverted ordering, or $\sum m_\nu > 0.12$ eV, refutes all three
F4	a massive graviton, a graviton dispersion violating $c = 1$, or a UV divergence in the lattice gravitational sector (the [Paper II Section 3] field: the [Paper II Section 2.4] quintet channel is not claimed to be the graviton, and its speeds $g_{10}\{1, 2\} \neq 1$ and its E_g mass shift are named defects of the channel)

#	observation that refutes the theory
F5	proton decay: the desert solution predicts $\tau_p \simeq 1.8 \times 10^{36}$ yr; observation of GUT-pattern decay far below that value (e.g. at Hyper-Kamiokande reach, 10^{35} yr) or non-observation beyond 10^{37} yr refutes it. At the closed angle the same content gives $\tau_p = 2.8 \times 10^{39}$ yr with the $(1, 2)$ threshold above M_X, so on that branch only a null result past 10^{37} yr bites and the Hyper-Kamiokande channel closes; the $M_{\rm GUT} = M_P$ reading ($\tau_p \sim 10^{47}$ yr) is abandoned
F6	a baryon asymmetry outside 10^{-11}–10^{-9}, or evidence that heavy-neutrino decays are in equilibrium
F7	a dark-matter particle with mass far from 118 GeV in the weakly-interacting thermal case; a direct-detection portal signal above 10^{-50} cm^2 (the named extension predicts 5×10^{-51}, and every bond-number portal strength is already excluded by the LZ-class bound 1.5×10^{-47} cm^2: predicted 2.5×10^{-45}–6.3×10^{-41} cm^2); or a detected non-neutral/decaying dark sector

The most accessible tests are **F3** and **F5**. KATRIN currently gives $m_\beta < 0.45$ eV [4], so the bare ladder value 0.146 eV survives a single-mass search, but the cosmological sum $\sum m_\nu < 0.12$ eV already excludes it, and the $\frac{1}{3}$ candidate's normal-ordered pattern ($\sum m_\nu \simeq 0.05$ eV, $m_{\rm lightest} = 0$) is inside the reach of the next generation of oscillation and $\beta\beta$-decay data. The desert solution predicts $\tau_p = 1.8 \times 10^{36}$ yr, about one decade beyond Hyper-Kamiokande's 20-year reach ($\sim 10^{35}$ yr): Hyper-Kamiokande can **refute** it by observing GUT-pattern decay anywhere at its reach, and an exposure passing 10^{36} yr that sees nothing refutes it too; the confirmation window belongs to the megaton era. **F7** has narrowed: the carrier is not in the implemented spectrum and the named extension is nearly decoupled, so the test is the mass coincidence itself.

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8. Conclusion

One loop factor organises four scales across thirty orders of magnitude, and the automaton now supplies the factor itself: a closed traced loop of length 10 in the gauged bond layer reproduces $1/(16\pi)$ to 3.6%, the crossing of the suppression is exactly the ladder length, and the contributing loops are enumerated exactly. The coefficient of the hierarchy is a product of measured numbers, $3F_{10}^{10} = 2.0881$ against 2.0765 observed, conditional on the one-split-node-per-family choice. The unification of the three measured couplings at $M_X = 1.8 \times 10^{16}$ GeV and $\alpha_X = 1/25$ (standard normalisation) holds only with desert content carrying two fitted thresholds; the automaton's own Wilson mechanism forces the excluded meeting $M_X = 8.9 \times 10^{14}$ GeV, rigidly, and the surviving solution requires a flavour-dependent bond layer with six fitted couplings that no on-site algebra element, no class invariant and no second-order bond number supplies. The dark-matter mass scale stands and its carrier does not exist inside the implemented spectrum — the bond-strength portal is excluded by direct detection; the neutrino ladder is in tension and its $\frac{1}{3}$ candidate is a built structure with a free junction amplitude; the baryon asymmetry fails Sakharov 3 at the model's own couplings, as a theorem. Two conventions, eleven fitted coefficients and one conditional coefficient stand between the root and the data; nine open problems are each pinned to a named quantity; seven falsifiers are stated, and the two most accessible — the neutrino mass sum and the proton lifetime — are testable within the current decade.

References

1. M. Jungman, M. Kamionkowski, and K. Griest, "Supersymmetric dark matter," *Phys. Rep.* **267**, 195 (1996).
2. LUX-ZEPLIN Collaboration (J. Aalbers *et al.*), "Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment," *Phys. Rev. Lett.* **135**, 011802 (2025), arXiv:2410.17036.
3. Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).
4. KATRIN Collaboration (M. Aker *et al.*), "Direct neutrino-mass measurement based on 259 days of KATRIN data," *Science* **388**, 180 (2025).
5. Particle Data Group (S. Navas *et al.*), "Review of Particle Physics," *Phys. Rev. D* **110**, 030001 (2024).
6. M. E. Machacek and M. T. Vaughn, "Two-loop renormalization group equations in a general quantum field theory. I. Proof of the equivalence theorem," *Nucl. Phys. B* **222**, 83 (1983); "III. Scalar model with

fermions," *Nucl. Phys. B* **249**, 70 (1985).
7. Super-Kamiokande Collaboration (K. Abe *et al.*), "Search for proton decay via $p \to \nu K^+$ using 260 kiloton·year data of Super-Kamiokande," *Phys. Rev. D* **90**, 072005 (2014).
8. A. D. Sakharov, "Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe," *JETP Lett.* **5**, 24 (1967).
9. R. Barbieri, P. Creminelli, A. Strumia, and N. Tetradis, "Baryogenesis through leptogenesis," *Nucl. Phys. B* **575**, 61 (2000); W. Buchmüller, P. Di Bari, and M. Plümacher, "Leptogenesis for pedestrians," *Ann. Phys. (N.Y.)* **315**, 305 (2005).
10. J. A. Harvey and M. S. Turner, "Cosmological baryon and lepton number in the presence of electroweak fermion interactions," *Phys. Rev. D* **42**, 3344 (1990).
11. S. A. Khlebnikov and M. E. Shaposhnikov, "The statistical theory of anomalous fermion number nonconservation," *Nucl. Phys. B* **308**, 885 (1988).
12. S. Davidson and A. Ibarra, "A lower bound on the right-handed neutrino mass from leptogenesis," *Phys. Lett. B* **535**, 25 (2002).
13. S. Blanchet and P. Di Bari, "The minimal scenario of leptogenesis," *New J. Phys.* **14**, 125012 (2012).
14. A. Pilaftsis and T. E. J. Underwood, "Resonant leptogenesis," *Nucl. Phys. B* **692**, 303 (2004).
15. Super-Kamiokande Collaboration (A. Takenaka *et al.*), "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in 0.37 megaton-years exposure of Super-Kamiokande," *Phys. Rev. D* **102**, 112011 (2020).
16. J. Hisano, D. Kobayashi, Y. Muramatsu, and N. Nagata, "Two-loop renormalization factors of dimension-six proton decay operators in the supersymmetric standard models," *Phys. Lett. B* **724**, 283 (2013).
17. M. Nath and P. Fileviez Pérez, "Proton stability in grand unified theories, in strings and in branes," *Phys. Rep.* **441**, 191 (2007).
18. H. Fritzsch, "Weak-interaction mixing in the six-quark theory," *Phys. Lett. B* **73**, 317 (1978).
19. I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou, "The fate of hints: updated global analysis of three-flavor neutrino oscillations," *JHEP* **09**, 178 (2020).
20. I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, "NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations," *JHEP* **12**, 216 (2024), arXiv:2410.05380.
21. P. F. Harrison, D. H. Perkins, and W. G. Scott, "Tri-bimaximal mixing and the neutrino oscillation data," *Phys. Lett. B* **530**, 167 (2002).
22. L. Wolfenstein, "Parametrization of the Kobayashi-Maskawa matrix," *Phys. Rev. Lett.* **51**, 1945 (1983).
23. S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989).
24. E. W. Kolb and M. S. Turner, *The Early Universe* (Addison-Wesley, 1990), Ch. 5.
25. L. J. Hall, K. Jedamzik, J. March-Russell, and S. M. West, "Freeze-in production of FIMP dark matter," *JHEP* **03**, 080 (2010).

Code and data availability. All the experiments, simulations and figure generators behind this paper are publicly available at https://github.com/MohammedHAOUACH/CL10. Every number quoted in the paper is produced by these open scripts and can be reproduced independently from the repository. The verification status is stated as the audit found it: the numbers were verified internally (spot-checks against the text, the second symbolic construction of [Paper I] Appendix A, the exact 3^n-word loop enumeration); the pipeline was not independently executed by a second party, and independent re-execution is invited.

Competing interests. The author declares no competing interests.